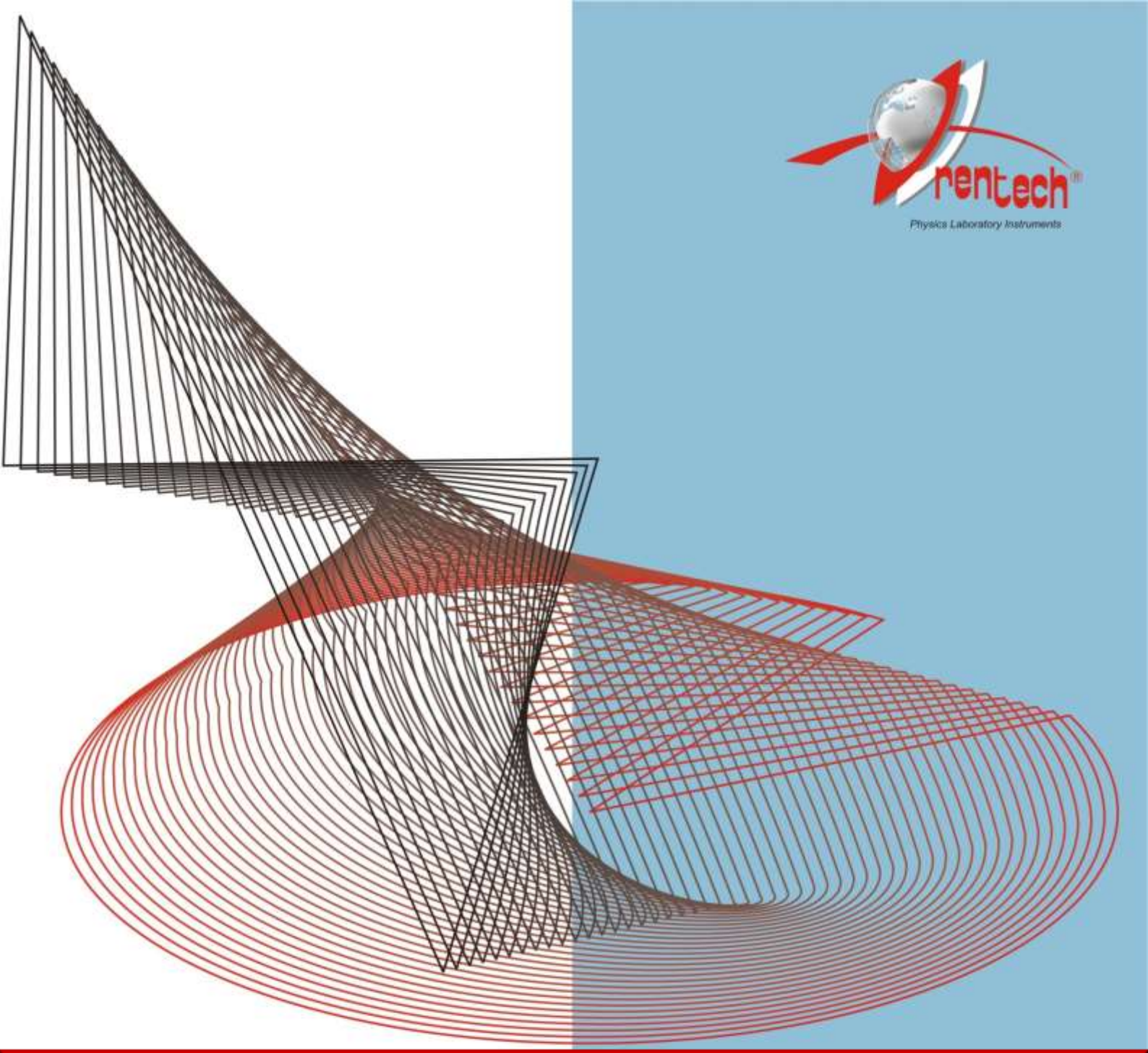


MEASUREMENTS, ERRORS AND GRAPHS

(Economic Version)

Student's Guide

Laboratory Manual and Workbook

Ankara-2015

Package List

1. A Set of Objects with Different Shapes
2. Digital Balance
3. Micrometer
4. Caliper
5. Graduated Cylinder
6. Ruler
7. A Set of Mass with Different Weights
8. Instructional Manuals (**Teacher and Student Guides**)

Note: This version of the product does NOT include the spherometer, the set of lenses with different focal points and the equal-arm balance unless specifically requested in the purchase order.They are included in the advanced version of the product.

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1. Purpose

The purpose of this experiment is to;

1. Learn how to use caliper and micrometer,
2. Measure external dimensions (length, volume) of a given regular body,
3. Determine depth of a given object using a caliper,
4. Determine the possible **errors** in the measurements by ruler, caliper and micrometer, and compare their **accuracy**,
5. Determine the density of a regular body of known mass using different measurement techniques.

2. Physics and Measurement

Physics is one of the most fundamental sciences. It deals with behavior and structure of matter. It is based on experimental observations and quantitative measurements. In order to identify meaningful relationships that characterize nature, it is important to know how to measure physical quantities in terms of which the laws of physics are stated.

Physical quantities are not generally independent to one another. Physics is composed of seven fundamental quantities. In mechanics, there are three fundamental quantities which are length, mass and time. The other quantities used in mechanics can be derived in terms of these three. The other four are; current, thermodynamic temperature, amount of substance and luminous intensity.

2.1. International System of Units (SI Units)

To make accurate and reliable measurements, a standard must be defined that do not change in various locations. Scientists and engineers all over the world use the same system of units as a standard to measure physical quantities.

This system is the International System of Units (SI Units). It is based on the metric system and basic set of units including meter, kilogram and second. The SI base units are given in the Table-(1) below.

Table-**1**: Fundamental quantities and their units

Base Quantitiy	SI Base Unit	Symbol
Length	**meter**	*m*
Mass	**kilogram**	kg
Time	**second**	s
Current	**ampere**	A
Thermodynamic Temperature	**kelvin**	K
Amount of Substance	**mole**	mol
Luminous Intensity	**candela**	cd

2.2. The Prefixes

In metric system, the larger and smaller units are defined in multiplies of 10 from the standard unit. Commonly used prefixes are listed in Table-(2).

Table-**2**: Commonly used prefixes

Power of ten	Prefix	Symbol
10^{-15}	femto	**f**
10^{-12}	pico	**p**
10^{-9}	nano	**n**
10^{-6}	micro	**μ**
10^{-3}	milli	**m**
10^{-2}	centi	**c**
10^{3}	kilo	**k**
10^{6}	mega	**M**
10^{9}	giga	**G**
10^{12}	tera	**T**

2.3. Types of Measurements

In general, measurements are divided into two groups, namely direct and indirect measurement. Direct measurements of a physical variable are obtained by the reading of the scale of an instrument calibrated in the corresponding units of the variable.

Example of a direct measurement is that measuring a certain length with a ruler. Indirect or calculated measurements are the deduction of the magnitude of physical variables using mathematical operations with magnitudes obtained by direct measurements.

In general terms, indirect measurements are those that correspond to physical variables whose values have not been determined by an instrument that can measure them directly. As an example, calculation of a sphere (of which radius,r, is measured initially) by the formula of $V = \frac{4}{3}\pi r^3$.

3. Errors

No physical quantity can be measured with a perfect certainty; there are always errors in a measurement. It is simply an observed fact that when independent measurements are repeated on a physical system, all the measured values will not generally be exactly the same. In general, the types of errors in a particular experiment may be due to the observer, experimental equipment or the design of the experiment. In recording a measurement, it is important to estimate and record the error. Error in a measurement can be indicated by making a reasonable assessment of the accuracy of the measurement in terms of the sources of errors and estimating the maximum possible error in the measurement. Then, the result of the measurement is written by the maximum possible error using the symbol "$\pm$" followed from the value observed from the scale. As an example, consider "x" is the actual measurement and "Δx" is the maximum reading error, then the result of the measurement is written as "$x \pm \Delta x$".

3.1. Types of Errors

There are two types of errors which are systematic errors and random errors.

3.1.1 Systematic Errors

The source of systematic errors is due to the imperfections of the measuring instruments. Systematic errors are fixed errors and for each measurement, this fixed error value is either subtracted or added.

For instance, when a voltmeter without its reset setting done shows 0.1 volts when not in use, then an excess value of 0.1 volts is taken into consideration in all measurements we conduct with the same voltmeter. If 0.1 volts are subtracted from all measurements, the correct measurement value is obtained. In carrying out an experiment, it is important to consider possible sources of systematic errors and to take precautions to eliminate them during the construction and calibration of the instruments.

3.1.2. Random Errors

These errors arise from unknown and unpredictable variations in the experimental situations. Random errors are generally very small and the chances of errors being positive or negative are equal. These random errors emerge due to the diversity of the mechanism and may not be completely removed (as mentioned in our example given for systematic errors) from the experiment results. These errors require statistical analysis because of their randomness. The best way to determine random errors is to repeat the same measurement more than once. However, we will use another method since it is not always possible to make repeated measurements: the error in a measurement device is equal to the smallest amount that device can measure.

Random Errors in Position Measurements

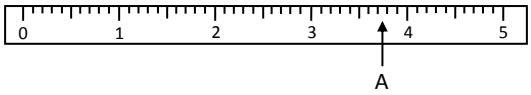


Figure-1: Measuring the position of A using a ruler.

Let us assume we are making a measurement with a ruler. The error in the ruler is equal to the smallest segment length on the ruler. For example, as seen in the **Figure-(1)**, we are measuring a position measurement of a point A with a ruler divided into "mm" segments and find the result to be 3.7 cm. A reading may appear in between points 3.7 and 3.8 cm.

Because the smallest quantitative value of our device can measure is 1 mm which is standard for all position measurements with this ruler, then, we write our result as $3.7 \pm 0.1\ cm$. Thus, we may mention that a ruler divided into millimeters is more reliable than one divided into centimeters.

Random Errors in Length Measurements

It is a measurement error in determining a distance between two points in one dimension.

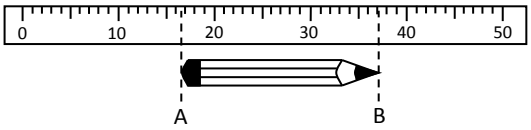


Figure-2: Measuring the length of a pencil using a ruler.

As an example, in the Figure-(2), let us measure the length of a pencil using a ruler. The length of the pencil is calculated by the difference between the end points of A and B. Therefore, the positions of these two points should be measured with respect to the same reference point. Each position measurements come with an error of 1 mm. Therefore, the error of this measurement is 2 mm. As seen in the figure, we can read the position of point A as 17 cm and position of point B as 37 cm with respect to the "0" point on the scale of the ruler. Then, we write our result as:

$$L \pm \Delta L = (37 - 17) \pm 0.2cm = 20 \pm 0.2\ cm.$$

As being experienced in this example, the effect of random errors can be reduced by improving the experimental technique and repeating the measurement many times and taking the average.

- ☑ The scale of a measuring instrument is not exact.
- ☑ The exact reading of a scale is impossible.
- ☑ The exact measurement of any quantity is impossible.
- ☑ No experiment gives the right answer.

3.2. Accuracy and Precision

Experimental errors are the difference between a measurement and the true value or between two measured values. Under the effect of these errors, the reliability of the measured values are analyzed by its *accuracy* and *precision*. Accuracy and precision are two important factors to consider when taking data from measurements. The accuracy of an experiment is a measure of how close the experimental result comes to the *true (or accepted) value*. It is related to a measure of the correctness of the result. Source of systematic errors determines how accurate the measurement is. The precision of an experiment is a measure of how reproducible the result is. That is, it is a measure of the magnitude of the uncertainty of the result due to random errors. It is related to a measure of repeatability of the result and therefore, it is usually limited by the sources of the random errors.

Both accuracy and precision reflect how close a measurement is to an actual value, but accuracy reflects how close a measurement is to a known or accepted value, while precision reflects how reproducible measurements are, even if they are far from the accepted value.

To clarify these definitions, consider an example about throwing and striking a bullet to a target (Figure-3).

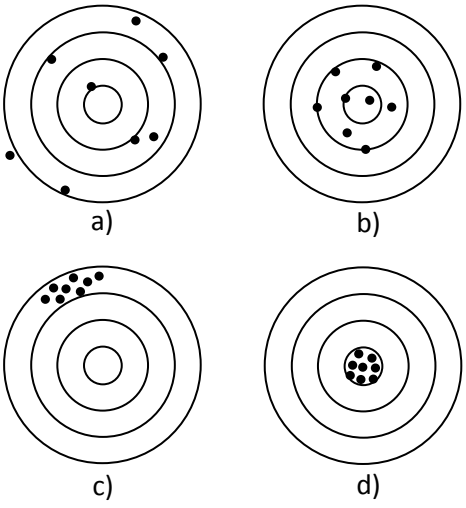


Figure-3: The difference between accuracy and precision.

In the **Figure-(3),** in the Part-(a), there is an example showing an example of striking which is neither precise nor accurate. Because the aim of throwing the bullets is to hit the target on its center, they are not in the middle of the inner circle and also the differences between them are very large. Part-(b) shows an example of striking which is accurate but not precise. As compared to the results in part-a, the bullets are near the centre of the target but they are not close to each other. Part-(c) shows an example of striking which is precise but not accurate. As seen in Part-(c), the bullets are in a group which means the results are approximately the same; the difference between the bullets is very low. However, they are not located near the centre of the target. Lastly, Part-(d) shows an example of striking which is both accurate and precise. This means that each of the trial gives the true (accepted) result, the bullets are located near the centre of the target, and also they give the similar results, the difference between the bullets is small.

The accuracy of an experiment often depends on both systematic and random errors. The precision of an experiment typically depends on random errors.

4. Significant Figures

When certain quantities are measured, the measured values are known only to within the limits of the experimental uncertainty. The number of significant figures in a measurement can be used to express something about the uncertainty. The number of significant figures is related to the number of numerical digits used to express the measurement.

The degree of certainty of a measurement is implied by the way the result is written or reported. While reading the value of an experimental measurement from a calibrated scale, only a certain number of digits can be obtained or read. Significant figures are those figures which are known to be reasonably trustworthy; they are figures actually obtained from the measuring instruments, including zeros and estimated figures.

The position of the decimal point does not affect the number of significant figures. If a zero is used to locate the decimal point, it is not significant, but if it actually represents a value that is read on the instrument, or estimated, then it is significant.

Give the figures you read from a scale with only the last figure estimated as a result of measurement. Never put extra figures. The digits including the last one estimated are called significant figures.

The number of significant figures in a measurement is determined as follows:

- ☑ The leftmost nonzero digit is the most significant.
- ☑ If there is no decimal point, the rightmost nonzero digit is the least significant.
- ☑ If there is a decimal point, the rightmost digit is the least significant, even if it is zero.
- ☑ All digits between the least and most significant digits are considered to be significant.

Pay attention to zeroes whether they are significant or not.

As an example, the numbers in the following list of numbers all have four significant figures. There is an explanation for each.

- ✓ 6739: All four nonzero digits are significant.
- ✓ 456900: The last two zeros are not significant because there is no decimal point.
- ✓ 0.008589: Zeros at the left are never significant.
- ✓ 0.01000: The zero at the left is not significant, but since there is a decimal point, the three zeros at the right are significant.
- ✓ 10.60: There is a decimal point, so all four numbers are significant.
- ✓ 1.057: Again, there is a decimal point, so all four are significant.

4.1. Scientific Notation

Physics deals with some very large and very small numbers. The scientific notation is a convenient method to express numbers rather than conventional notation. Scientific notation also allows numbers to be expressed in a form that clarifies the number of significant figures. The exponent in scientific notation is equal to the number of times that the decimal point must be moved to produce a number between 1 and 10.

Write the result of a measurement as two factors; the first factor contains all the significant figures, having one nonzero digit in front of the decimal point, and the second factor is a power of 10.

Examples for the usage of this notation are given below;

- ✓ $255\,mm = 2.55 \times 10^2\,mm$
- ✓ $11.1\,cm = 1.11 \times 10^1\,cm$
- ✓ $0.0125\,km = 1.25 \times 10^{-2}\,km$
- ✓ $360\,g = 3.60 \times 10^2\,g$

4.2. Calculation with Significat Figures

The calculations carried out with significant figures usually produce extra figures as a result of calculation. It is important that the calculations should be consistent with the values that have measured. The value with the least significant digits sets the standard for the degree of accuracy for the measurements. This means that the result calculation should have only one estimated figure.

4.2.1. Rounding Numbers

The result of the measurements can be rounded off to the desired number of significant figures before they are used in calculations. The rules for rounding of the numbers are:

- ☑ If the last figure is less than 5 (0, 1, 2, 3 or 4), it should be dropped and the others should be unchanged.
- ☑ If the last figure is equal to 5 or greater (5, 6, 7, 8 or 9), it should be dropped and the numerical result should be increased by 1.

For instance,

- ✓ 2.431 may be rounded off to 2.43 when rounded off to the nearest hundredths place.
- ✓ 2.431 may be rounded off to 2.4 when rounded off to the nearest tenths place.
- ✓ 2.431 may be rounded off to 2 when rounded off to the nearest units place.
- ✓ 6.587 may be rounded off to 6.59 when rounded off to the nearest hundredths place.
- ✓ 6.587 may be rounded off to 6.6 when rounded off to the nearest tenths place.
- ✓ 6.587 may be rounded off to 7 when rounded off to the nearest units place.

4.2.2. Multiplication and Division

In the multiplication or division of the numerical results, the result should be rounded off to the least number of significant figures used in the calculation. For example,

```
   3.519                          3.5 (rounded off)
 x 2.3 (least significant)      x 2.3
 -------                        -----
   10557                          105
+ 7038                         +  70
 -------                        -----
 8.0937 (≈8.1)                  8.05 (≈8.1)
```

As can be seen from the example, since the least number of significant figures is 2, the result is rounded off 2 significant figures.

The same result can be obtained by rounding off the all multipliers to the least number of significant figures before the multiplication.

$$\frac{102.35}{15.72} \Rightarrow \frac{10235}{-9432}\,\Big|\,\frac{1572}{6.5108\ (\approx 6.511)}$$

$$8030$$
$$-7860$$
$$1700$$
$$-1572$$
$$12800$$
$$-12576$$
$$224$$

The example above shows a division of two measured values. Similar to the multiplication case, the number of significant figures of the result should be equal to the number of significant figures of the value with the least number of significant figures. Therefore, the result has 4 significant numbers.

In the multiplication or division of numerical measurements, retain in the result only as many figures as the number of significant figures in the least accurate number used in the calculation.

4.2.3. Addition and Substraction

When the numerical results are added or subtracted, the answer can contain no more decimal places than the least accurate measurement. This means that the number of places after decimal point in the answer should be less than or equal to the number of decimal places in each term in the calculation. For example,

$$\begin{array}{r} 0.073 \\ 13.7 \\ +\ 156 \\ \hline 169.773\ (\approx 170) \end{array}$$

The same result can be obtained by rounding off the all terms to the least accurate terms in the calculation before addition.

When carrying out addition or subtraction with numerical measurements, do not carry the result beyond the first column (from the left) that contains a doubtful digit.

5. Errors in Calculated Results

In the real case measurements, obtained physical quantity should be written with errors of the measurement. Whenever the measurements are used in calculations, the error should be used in these calculations to move each uncertainty of the measurements to the final result.

To work on the usage of errors in the calculations, assume that there are two recorded measurements as $x \pm \Delta x$ and $y \pm \Delta y$.

5.1. Addition and Substraction

In these calculations, the maximum possible error ΔR in the final result R is the addition of errors from these measurements.

For addition:

$$R \pm \Delta R = (x \pm \Delta x) + (y \pm \Delta y) = (x + y) \pm (\Delta x + \Delta y)$$

For subtraction:

$$R \pm \Delta R = (x \pm \Delta x) - (y \pm \Delta y) = (x - y) \pm (\Delta x + \Delta y)$$

When measured quantities are added or subtracted, the maximum possible error in the result is the sum of the errors in the quantities used in the calculation.

5.2. Multiplication and Division

The multiplication and division of two measured data is a little complicated than the previous calculations. They are performed as follows:

For multiplication:

$$R \pm \Delta R = (x \pm \Delta x).(y \pm \Delta y) = x.y \pm x.y\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

In this equation, the first term is the product of the actual measurements and second term is its corresponding error.

For division:

$$R \pm \Delta R = \frac{(x \pm \Delta x)}{(y \pm \Delta y)} = \frac{x}{y} \pm \frac{x}{y}\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

In this case, the first term is the division of the actual measurements and second term is its corresponding error.

6. Expressing Errors (Statistical Overview)

6.1. Average (Mean) Value

As mentioned in the types of errors part, random errors can be minimized by repeating the measurements several times. It is expected to obtain similar results for all trials. For this set of measurements, the actual result is most probably obtained by the average value. The average or mean value of a set of n measurements can be expressed as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

6.2. Average Deviation of the Mean

Taking the average value of a series of measurements due to the random errors, there is a difference between each result of the measurements from the average value. This deviation d_i of any single measurement x_i is written as:

$$d_i = x_i - \bar{x}$$

When using the average value of the results, the best estimated error is defined as the average deviation of the mean (A.D.):

$$A.D. = \frac{d_1 + d_2 + d_3 + \cdots + d_n}{n}$$

6.3. Standard Deviation

The precision of the measurements can be determined by calculating a quantity called standard deviation from the mean of the measurements. The standard deviation represents the average of the square of the variation of individual measurements about the average value. It is defined as:

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \cdots + d_n^2}{n}}$$

6.4. Absolute, Fractional and Percentage Errors

The error in a measured quantity may be expressed either as the absolute error, the fractional (or relative) error or percentage error. The absolute error is defined as the magnitude of difference between actual and measured value and it has the same units as the measured quantity. For example, a distance measurement made with a ruler marked in millimeters can be expressed as $25.5 \pm 0.2\,cm$. The absolute error in this quantity is $\pm\, 0.2\,cm$.

Fractional error is the ratio of the absolute error to the quantity and it has no unit. It can be expressed as:

$$Fractional\ error = \frac{Absolute\ Error}{Actual\ Value}$$

It is commonly expressed as a percentage error of a measurement to give the error to the actual value multiplied by 100.

$$Percentage\ Error = Fractional\ Error \times 100\%$$

The number of decimal places of the error part should be same with those of measurement.

For example,

- 25 ± 1 (true)
- 13.25 ± 0.05 (true)
- 25.2 ± 0.12 (false)
- 12.5 ± 0.5 (true)

7. Graphs

Graphing is a way of displaying relations between various quantities, parameters, or measurable variables in the measurements for clarity. Basically, it is a visual picture of the quantitative relationship between the physical quantities defining how one quantity changes if another quantity that is related to it also changes in the measurements.

Generally, in basic physics experiments, two-dimensional graph is enough to represent the relation between two related variable physical quantities. In these types of graphs, the horizontal axis (*x-axis* or *abscissa*) is used for the *independent variable*, and the vertical axis (*y-axis* or *ordinate*) is used for the *dependent variable*. For example, in a free fall experiment, assume that the measurement is done to determine the height *h* from the ground of a ball falling freely. In this case, the dependent variable *h* is investigated with the independent variable time *t*.

Moreover, in all plotting procedures, one of the most important points is to define the location of the data points obtained from the measurement according to the reference point called as the origin. It means that, any data point in the two-dimensional graph can be symbolized as (*x, y*) which refers to its coordinates on the *x* and *y* axis, with respect to the origin (0, 0).

Because the aim of usage of the graphs is summarizing a lot of information into one picture, a graph should be self-explanatory. The guide lines of graphs expecting from the experimental works are:

- Write the **title** of the graph
- Show x and y **labels** clearly
- Specify **units** on the axis
- **Scale** the axis correctly
- Identify the **data points** clearly
- Show the **error bars**
- Draw **a best** and **a worst line**
- Show equation of the **slope** and **y-intercept**

Title of the graph

- Each graph should have a title. The title is related to the variables that will be plotted. It is very important that someone should figure out easily what is plotted. A title is usually written for example “Distance (**D**) versus Time (**t**)” or “Voltage (**V**) versus Current (**I**)”.

Label and the Unit of the Axes

- ***x*** and ***y***-axis should be labeled according to the experimental data. Labels on the axes should have a unit. For instance, for the ***x***-axis which is used for the independent variable can labeled as “Time (seconds)” and ***y***-axis can “Distance (meter)”.

Scaling of the axis

First of all, the data numbers should be observed in order to determine how big a graph to make. Scaling should be appropriate so that the data points are distributed as widely as possible over the area of the graph paper. The independent variable lies on the *x*-axis and dependent variable lies on the *y*-axis. The data values are examined separately to fit the axes. The scalings may be different on each axis and does no necessarily need to start from zero. Each axis must be divided into equal parts in itself. The starting and the end points of the axis are determined by subtracting the error part from the minimum actual data and adding the error part to the maximum actual data. This method is done for each variable and start points from the origin can vary.

Identifying the Data Points

Each row of the data table indicates a data point. If there are five data values for each dependent and independent variable, then five data points should be identified on the graph.

Error Bars

Error bars represent the uncertainity of each data points. While drawing the error bars on the data points, the scaling on each axis is considered and the calculations for the error bars should be similar to the scaling calculations.

Up to these steps, the graph should be similar to the one given below in the **Figure-(4)**.

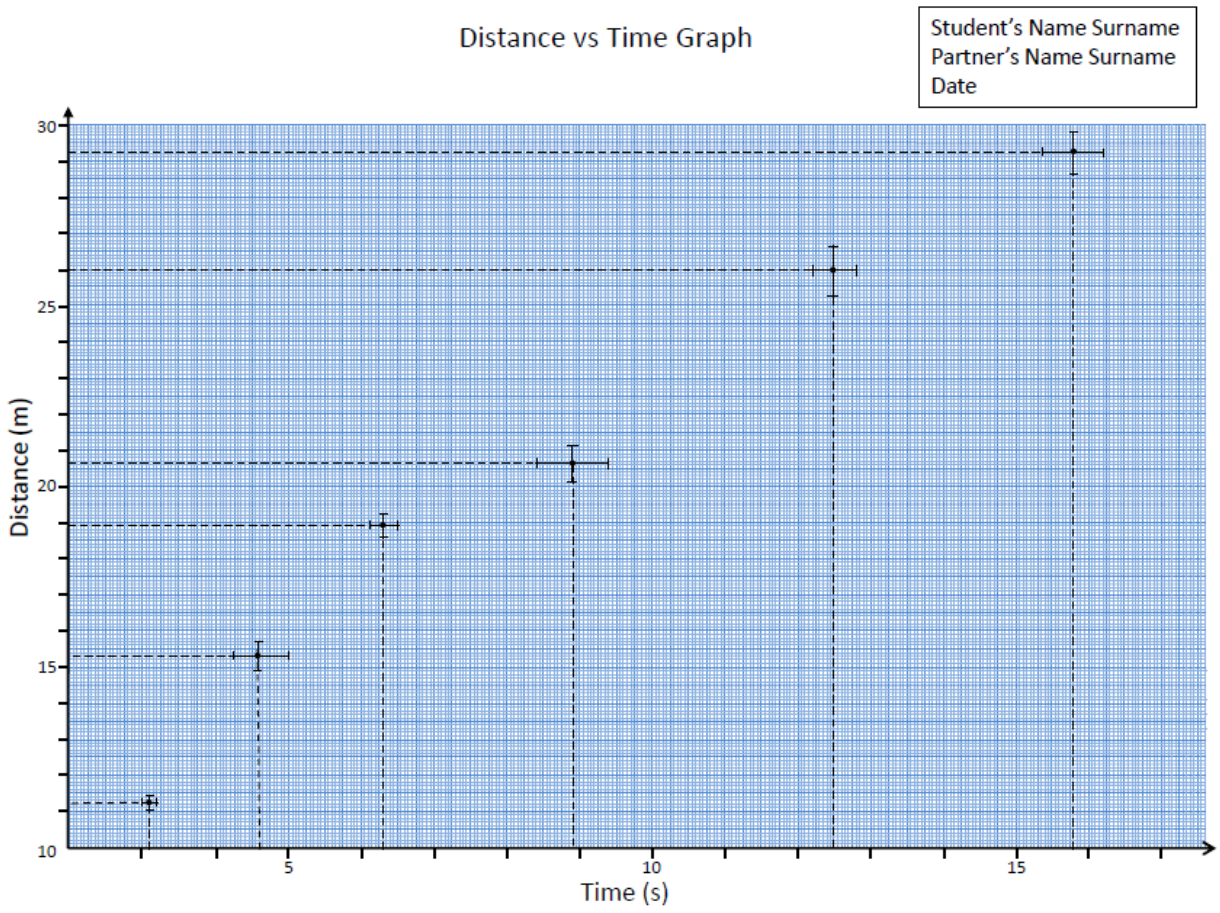


Figure-4: A graph of Distance vs Time.

7.1. Analysis of a Straight Line Graph

Two linearly related quantities, x and y, have an algebraic equation of the form:

$$y = mx + b$$

where m and b are constants. $\boldsymbol{y}$ vs $\boldsymbol{x}$ graph gives a straight line with slope m, which usually corresponds to a physical quantity in the measurements.

Assume that there are two points (x_1, y_1) and (x_2, y_2) in a data set of a measurement that shows a straight line behavior when plotted using $\boldsymbol{x}$ and $\boldsymbol{y}$ axis (Figure-4).

The changes in x and y axis between these points are defined as $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$, respectively.

Slope, $\boldsymbol{m}$ is expressed as;

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

In slope calculations, only one data point on the line cannot be chosen. Determining any arbitrary two points on the line is enough to calculate the slope. Because measured points contain an error part, we are required to draw a best straight line fitting to the data points to calculate the best value of the slope m. A best line is a straight line that best represents the data on the graph.

This line may pass through all, some or none of the data points. The main important point is that an equal number of points lie on each side of the best line.

The best straight line, however, must pass inside as many error bars as possible.

In order to calculate the error on ***m***, we need a line that worst fits to the data points. It must be chosen to give the maximum possible deviation from the best line. The drawn line must pass through the **error bars** of the first and last data points and the worst line must cross the best line.

Best and worst lines cannot be parallel.

The ***b*** in the equation is called the **intercept** and it gives the point at which the straight line crosses the *y*-axis. Simply, when $x = 0$, then $y = b$.

Actually in the analysis of straight line graphs, the aim is to determine from the graph the slope as $m \pm \Delta m$ and the intercept as $b \pm \Delta b$, where Δm and Δb are the maximum possible errors in slope *m* and in intercept *b*, respectively.

These errors can be easily determined by drawing the *worst possible straight line* on the graph. The analysis of this worst possible line is also done in the same way to obtain the worst possible slope *m'* and the worst possible intercept *b',* and then we get the possible errors Δm and Δb as:

$$\Delta m = |m - m'|$$

and

$$\Delta b = |b - b'|$$

After drawing the best and worst lines, the calculation steps of slope of best and worst line should be shown on the graph as in the example given below in the **Figure-(5)**.

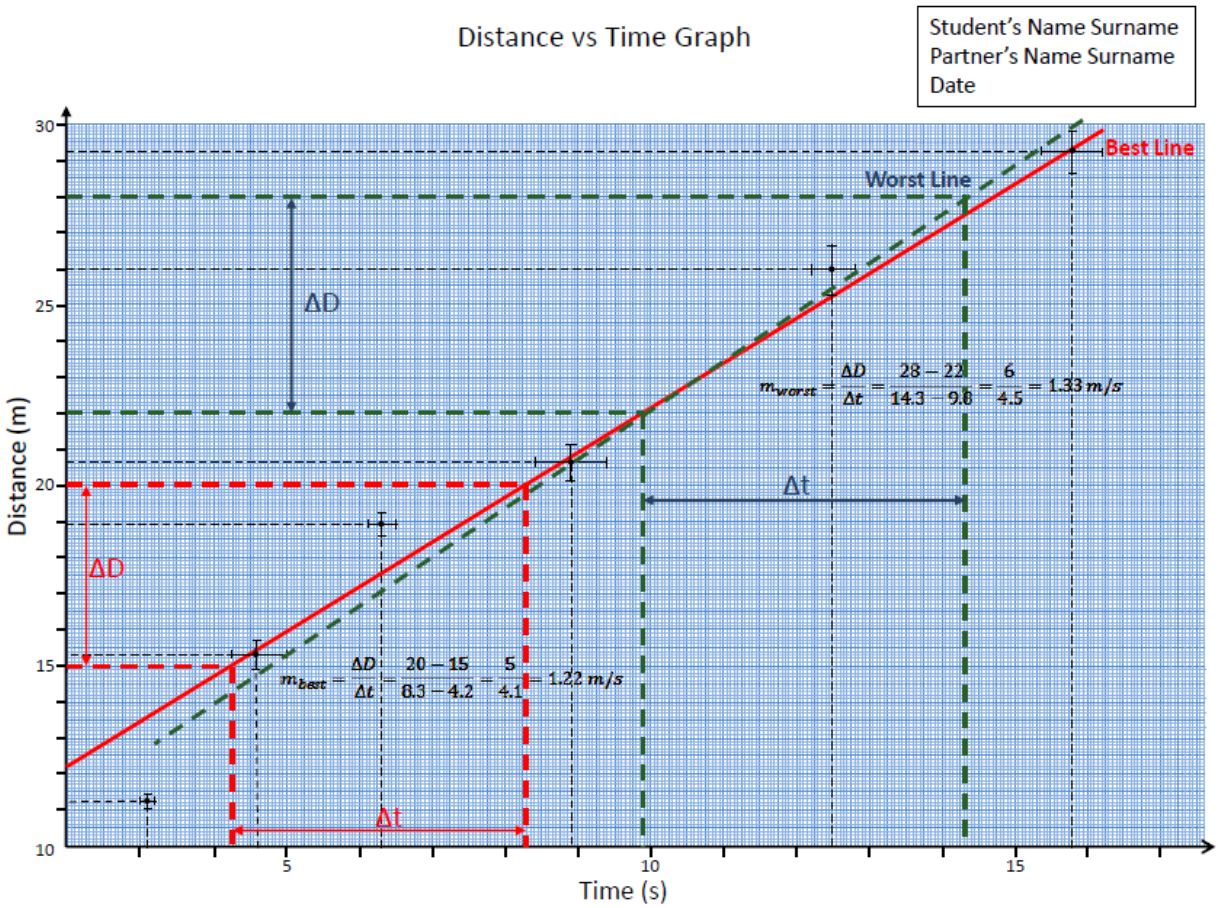


Figure-5: Slope calculations on the graph.

7.2. Nonlinear Graphs

In nature, the simplest relation to carry out in the experiments is direct relation as stated in the straight line analysis. However, the other relations are often valid in our daily life.

7.2.1. Inverse Proportion

This graph states that x is inversely proportional to y. This inverse relation can be rewritten as:

$$yx = k$$

or,

$$y = k/x$$

where $\boldsymbol{k}$ is a constant.

In such a relation, when $\boldsymbol{y}$ versus $\boldsymbol{x}$ graph is plotted, it gives a hyperbolic relation as shown in the **Figure-(6)**.

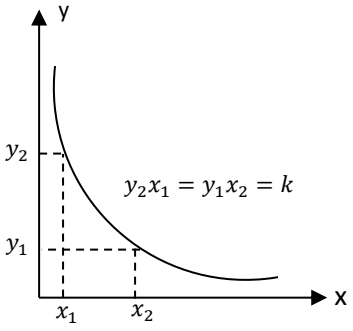


Figure-6: $\boldsymbol{y}$ versus $\boldsymbol{x}$ graph.

However, if the x-coordinate, independent variable is chosen as $1/x$, and then, if $1/x$ versus y graph is plotted (Figure-7), the direct proportion can be obtained.

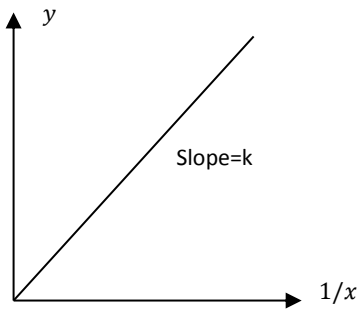


Figure-7: y versus $1/x$ graph.

Therefore, the experimental data can be studied as to be straight line graph.

7.2.2. A Square Law Graph

A square law relationship between any two variable *x* and *y* can be given as

$$y = kx^2$$

where *k* again a constant. In this relation, ***y*** is related non-linearly to the ***x***. Note that, instead of working on ***y*** versus ***x*** graph (Figure-8), it is also convenient to use a graph of *y* versus x^2. This graphical representation gives a straight line whose slope would be ***k*** as shown in the Figure-(9) and it would pass through the origin.

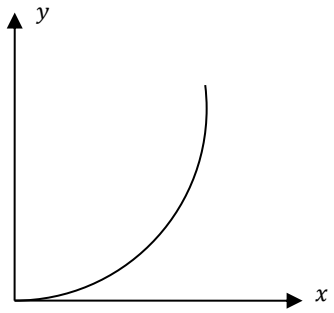


Figure-8: y versus x graph.

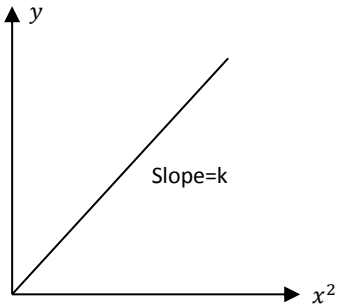


Figure-9: y versus x^2 graph.

7.2.3. An Inverse Square Law Graph

An inverse square law can be described as

$$y = k/x^2$$

where ***k*** again a constant. In this relation, the variable *y* is depending nonlinearly on ***x***, but it is depending linearly on $1/x^2$. Therefore, if the *x*-axis is chosen as $1/x^2$, the corresponding plot gives a straight line with a slope of ***k***.

7.2.4. Exponential Relation

If a graph of ***x*** and *y* has an equation given by

$$y = ke^{cx}$$

where ***c*** and ***k*** are some constants and *e* is the base of natural logarithm and equal to 2.718, taking the logarithm of both sides of this equation gives a straight line relation as

$$lny = lnk + cx$$

When lny is plotted as a function of x, the obtained straight line graph gives the slope as $\boldsymbol{c}$ and the intercept lnk (see **Figure-10**).

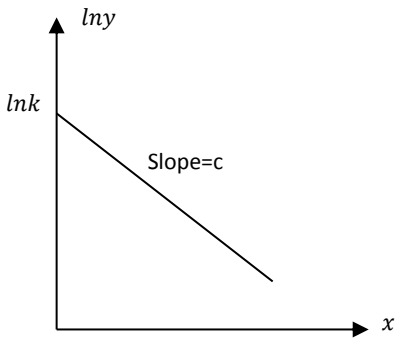


Figure-10: lny versus x graph.

Instead of using natural logarithms to the base 2.718, logarithms to the base 10 can also be used to get a straight line graph. The corresponding equation can be written as

$$logy = logk + cx$$

and the graph $logy$ versus x gives the slope as $\boldsymbol{c}$, and the intercept as $logk$.

7.2.5. Power Laws

Two variables $\boldsymbol{x}$ and $\boldsymbol{y}$ are related by the general equation,

$$y = kx^c$$

where $\boldsymbol{k}$ and $\boldsymbol{c}$ are constants. When taking the logarithm of both sides, the equation becomes

$$logy = logk + clogx$$

where it is in a straight line characteristics.

When $logy$ is plotted as a function of $logx$, obtained straight line gives the slope *c* and the intercept $logk$.

Instead of using logarithms to the base 10, natural logarithms to the base 2.718 can also be used to get a straight line graph. The corresponding equation can be written as

$$lny = lnk + clnx$$

and the graph lny versus lnx gives the slope as $\boldsymbol{c}$, and the intercept as lnk.

8. Experimental Set-up



Figure-11: Aluminum protective case for the measuring instruments.



Figure-12: The experimental set-up.

8.1. Ruler

A **ruler** is one of the most common measuring instruments to measure distances and to draw straight lines. Depending on the experimental requirements, there are many rulers with different sizes and shapes.

When measuring the length of an object, the measurement should not be taken from zero on the ruler, non-zero origin can also be chosen as a starting point. Reading a value from a ruler involves lining up the two points (one of the end point on the object and the reading point on the ruler) with the marks on the ruler, however, exact lining up is impossible. The apparent distance between two points on the object, and therefore the values of the reading, depends also on the position of your eye. Note that there are divisions on the ruler, and the *smallest* division is one millimeter. So, you should make a best *estimation* on a reading between the millimeter markings on the ruler. The position of a point near a ruler may appear different if viewed from the left or right of a line of sight perpendicular to the ruler, named as *parallax*. For example, thediagram shows the position of an object near the ruler.

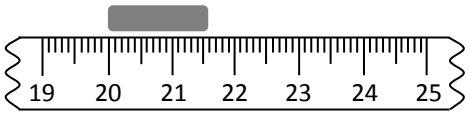


Figure-13: Measuring the length of an object.

Using the position of the ruler as shown in the **Figure-(13)**, the two edge points of the object should be estimated. While doing the measurements, be aware of the thickness of the millimeter lines on the ruler and the parallax effect on your observation. Therefore, for a one millimeter smallest division on the ruler, the estimation limits of error is equal to itself (1 mm). Then, the positions of the two end points of the object on the ruler can be written as 20.0 and 21.6 cm. Note that, the result of the measurements must be written with the correct number of significant figures and the unit of measurement.

As a difference between these two ends, the length of the object can be recorded as:

$$d = 1.6 \pm 0.1\ cm \text{ or } d = 16 \pm 1\ mm$$

The significance of $\pm\ 1\ mm$ (or $\pm\ 0.1\ cm$) is that when we repeat our measurement several more times, the readings we obtain are expected to be one of the values 15.0 mm, 16.0 mm or 17.0 mm. For many repetitions of the measurement, the value most frequently occurring will be the value 16.0 mm.

8.2. Caliper

A (vernier) caliper is a common tool used in laboratories and industries to accurately determine the fraction part of the least count division. Unlike a ruler, the vernier caliper allows the fractional part of the smallest division to be accurately determined, not merely estimated. It is a standard method used in instruments to get higher precision readings. It can be used to measure length of an object, the outside/inside diameter of a hole and the depth of a hole.

The schematic diagram is illustrated in **Figure-(14)**. As seen in this figure, the main parts of a vernier caliper are a main scale, a ***Vernier*** scale and a fixed and sliding jaw.

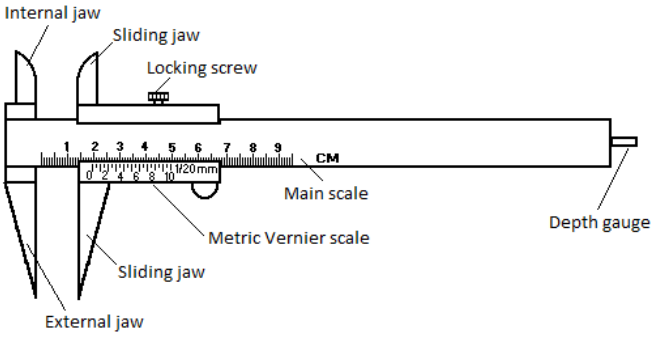


Figure-14: ***Vernier*** caliper.

Vernier calipers are available with metric and imperial graduations. Also, there are different types of caliper may employ other types of readout such as, a dial for least count readout or an electronic digital readout.

Calipers are available in various sizes, with measuring ranges from 100 mm to 3000 mm. Generally, calipers that have a measuring range of 300 mm or less are classified as small calipers and ones with larger ranges are classified as large calipers. Typical measurement resolution for a vernier caliper is 0.05 mm. This uncertainty in a measurement is changed depending on its type; it is 0.02 mm for dial calipers and 0.01 mm for digital calipers.

Using a caliper

The vernier consists of a main scale engraved on a fixed ruler and an auxiliary scale engraved on a moveable jaw (see **Figure 14**). The moveable jaw is free to slide along the length of the fixed ruler. The main scale is calibrated in centimeters with the smallest division in millimeters. The moveable auxiliary scale has 10 divisions that cover the same distance as 9 divisions on the main scale. Therefore, the length of the auxiliary scale is 9 mm. When the vernier is closed and properly zeroed, the first mark (zero) on the main scale is aligned with the first mark on the auxiliary scale. The last mark on the auxiliary scale will then coincide with the 9-mm mark on the main scale.

External Measurement

These types of measurements are the most basic usage of the calipers. In this measurement, at first, the jaws should be opened enough to place the caliper over the object to be measured, and then the movable jaw should be slid toward the fixed one until it contacts the object. Therefore, to measure length, place the object between the jaws of the caliper and lightly close the jaws on the object.

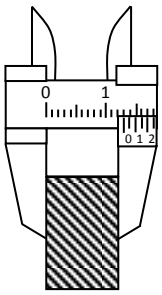


Figure-15: Measuring the length of the object.

However, there may be some errors that caused the result not to be accurate. The most effective source of error is occurred when the object is in contact with the jaws such as the characteristics of the objects (whether it is soft or hard), or your pressure to put the object between the jaws. Moreover, if the calipers are not straight (that is, the jaws are not in contact with the contact sides of object) or if there is dirt on the jaws or the contact points, the measurement may not be accurate.

Internal Measurement

To determine the internal diameter of a piece of object as shown in the **Figure-(16)**, the smaller jaws on the top of the caliper are used. In this measurement, at first, the jaws should be closed and placed into place to be measured. Then, taking the internal jaw is fixed at one end, movable jaw should be moved until it contacts with the other end point of the object. The locking screw should be tightened so that an accurate measurement can be made.

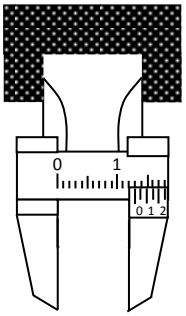


Figure-16: Measuring the inner diameter of the object.

In such type measurements, the most common problem is keeping objects lined up correctly when you are taking an inside measurement.

Depth Measurement

For the ***depth*** measurements, the depth gauge of the caliper is used. When sliding the movable jaw, this part extends out of the far end. In this measurement, initially one end of the caliper should be placed against top of the hole as shown in the **Figure-(17)**.

Then, using movable jaw, the caliper should be opened by sliding the movable jaw towards the hole until the depth gauge contacts the bottom of the hole. The locking screw is tightened and the measurement can be read on the scales of the caliper.

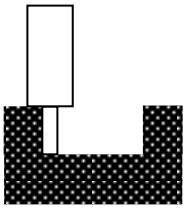


Figure-17: Measuring the **depth** of the object.

Measuring the **depth** of a hole by a caliper is difficult than the other types of measurements. Since caliper's depth gauge must be forced into contact with the bottom of the hole, in some cases it is difficult to determine the contact point and also to contact with the bottom point accurately.

Reading Calipers

On the caliper, there are two scales, a main (or fixed) scale and a Vernier (or sliding) scale. In typical metric calipers, the numbers on the main scale are given in centimeters and therefore the smallest division on that scale is a millimeter. Moreover, the Vernier scale is divided to 10 points from 0 to 10. Note that, when reading the Vernier scale, you start from the 0 on the sliding scale.

After the measurement is done, at first, the position of 0 point of the Vernier scale on the main scale is read. It is a rough estimation of the length of the measuring part of the object. Then, this **0** point should be observed whether it is in the same line with the mark on the main scale or not. If it is in the same positioning, the reading from the main scale can be taken as the result of the measurement with the instrumental uncertainty. On the other hand, if not, the line on the Vernier scale aligns with a line on the main scale must be found to get a correct reading result. This observed point must be evaluated considering the Vernier scale contains a hundred points.

In this case, the point is read on this scale in tens digit. This gives a reading for the hundredths place from the decimal point of the main scale reading. Again, the result of the measurement should be written with expressing the instrumental uncertainty. The following examples on an integer and non-integer readings explains how a reading is taken using a metric vernier caliper.

The **Figure-(18)** below shows the case where the zero graduation on the vernier scale is aligned perfectly with a graduation on the ***main scale***. In this case, the zero on the Vernier scale and the 35 mm graduation on the main scale is aligned. The correct caliper reading is therefore 35.00 mm.

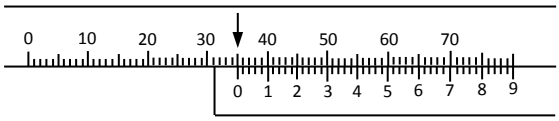


Figure-18: An integer reading on the main scale.

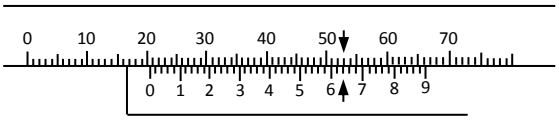


Figure-19: A non-integer reading on the main scale.

Figure-(19) illustrates the case where the ***zero*** on the Vernier scale does not line up with a graduation on the main scale. In this case the procedure for finding the correct reading is as follows.

1. Look along the main scale and determine the smaller of the two graduation numbers that most closely line up with the zero on the vernier scale. In the example above figure, the zero on the Vernier scale lies between 20 and 21 on the main scale. The correct main scale reading to use is therefore 20 mm.

2. Now look along the vernier or sliding scale and see which graduation lines up with a main scale division. In figure, this is 54 on the vernier scale, and note that the resolution of this scale is 0.01 mm.

3. The correct reading in the above example is therefore 20 mm + 54 x 0.01 mm = 20.54 mm.

Note that there is no ambiguity in the reading of the caliper so the reading uncertainty is zero. However, there is an instrumental uncertainty of ± 0.01 mm.

Taking several measurements on any one dimension and recording the mean can give extra confidence in measurements and is necessary for estimating the uncertainty of measurement.

Sources of the Errors

There are a number of factors affecting the measuring accuracy of calipers. The user should be aware of both the cause and effect of these errors. As an error source to the construction of the caliper, straightness of the caliper jaws are important to control before usage. These inside and outside measuring faces should be perfectly flat and also be parallel to one another.

Moreover, resolution of the Vernier scale is the other source of error that affects the accurate reading from caliper. As in the ruler usage, because in most of the measurements, the main scale and Vernier scale of a caliper are not in the same plane, the apparent alignment position can vary depending on the parallax error. Most of the calipers are not provided to a constant measuring force to the measuring object during the measurement. Therefore, the measuring force will vary each time when a measurement is made, especially with different users.

In addition to these, before taking a measurement, the surfaces of jaws and depth gauge, and also the measured faces of the object should be cleaned to eliminate measurement errors.

8.3. Micrometer

A micrometer, similar to a caliper, is a measuring instrument for precise measurement of a very small objects and/or distances. This device is based on a graduated screw mechanism to produce precise linear displacement of a spindle along its axis.

Distance measurement is achieved referencing the linear displacement of the spindle to a fixed measuring face on the axis of the spindle (the anvil). Figure-(20) shows the main components of a micrometer.

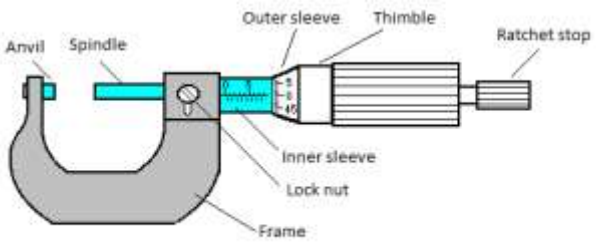


Figure-20: Components of the micrometer.

Like calipers, micrometers are generally used to measure the outside and inside diameters, and also the depth of the holes in an accurate way. In a micrometer usage, spindle and thimble reading lines are the parts enable to get reliable results in detail. A standard micrometer has a screw thread of 0.5 mm pitch with a thimble graduated in fifty equal divisions around its circumference. There are many micrometers that produced in different size ranges. In a typical way, as a smallest one 0 – 25 mm size range is available, and the others are formed in a 25 mm increment in size ranges. The choice is dependent on user's needs.

In order to get a precise measurement, the most effective case in micrometer usage is the force that spindle applies to the measuring object. The applied force varies the measurement result due to the deformation on the spindle or spring in a micrometer even if it does not affect the surface of the object. Therefore, a constant and same force should be applied to the object to minimize the variation in measurements and also prevent instrumental errors. In a micrometer, the ratchet stop is commonly used to facilitate this setting. The mechanism comprises two ratchets that are opposed to each other and whose toothed faces are pressed together by the force of a spring. When the barrel of the ratchet stop is rotated clockwise, both ratchets turn together until the measuring force reaches to a certain limit.

As the force exceeds this limit, the ratchet in the barrel spins idly and clicks. When the barrel is rotated counter-clockwise there is no idling between the ratchets because the ratchet teeth remain engaged and the thimble is allowed to rotate.

Reading a micrometer

To perform a measurement with a micrometer the initial procedure is that the anvil should contact with the surface of the measuring object. Then, with rotating the thimble or the ratchet until the spindle touches the object, the anvil and spindle should be close to (but not in contact with) the object.

The ratchet should be rotated until you hear a click from the ratchet stop to apply a pressure to keep the object in between anvil and spindle.

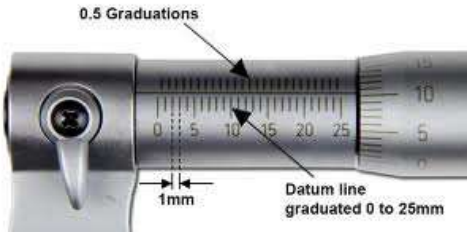


Figure-21: Reading a value from the sleeve scale of a micrometer.

In reading a measurement result on a micrometer scale, at first, read the number on the major divisions of the sleeve scale. Then, a further half mm measurement can be seen on the bottom half of this scale. After these two readings, the thimble scale must be read to complete the reading procedure. In this final step, the value of the line on thimble scale that coincides with the datum line on the sleeve is added to the sleeve scale reading. With 0.01 mm discrimination micrometers, there is an instrumental uncertainty of $\pm\, 0.01\ mm$.

An example of ***reading*** a micrometer is shown in the Figure-(22).

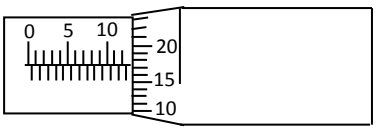


Figure-22: Micrometer reading.

1. The major sleeve scale reading is 12 mm.

2. A further half mm measurement seen on the bottom half of the scale gives an addition of 0.5 mm. Therefore, measurement reading on the sleeve scale is 12.5mm.

3. The line on the thimble scale coincides with the datum line shows 16 full divisions (these are hundredths of a mm). Therefore, the correct reading in the above example is 12.5 mm + 16 x 0.01 mm = 12.66 mm.

4. Finally, the uncertainty in the micrometer reading should be added to the result as 12.66 ± 0.01 mm.

Taking several measurements on any one dimension and recording the mean can give extra confidence in measurements and is necessary for estimating the uncertainty of measurement.

Sources of Errors

At first, it is important for the user to be aware of the possible error sources in the measuring instrument, micrometer. The calibration and mechanical facilities should be checked before usage. Moreover, the surface characteristics of the measuring object affects the final result. It is because of the variations in contact points of object surfaces with anvil and spindle. The accuracy in result is obtained by the thimble scale reading. Since the reading is somehow estimation, the minimum graduation on this scale determines the uncertainty of the result.

8.4. Graduated Cylinder

Graduated cylinders are used to measure volumes accurately. It can be used to calculate the density of an object if its mass is known. Graduated cylinders usually are made from plastic or borosilicate glass. Common sizes are 10, 25, 50, 100, 250, 500, 1000 ml.

Water Immersion Method for Volume Determination

The volume of a solid object can be measured by using the water displacement method. In this method, the object is placed in a liquid that is less dense than the object. The addition of the object to the liquid raises the level of the liquid because the two volumes are additive.

This experiment assumes that the solid doesn't react with or dissolve in the liquid. V_1 is the initial volume of the liquid. V_2 is the sum of the volume of the liquid and the solid object. Notice that V_2 is greater than V_1. This is because the volume of the liquid and the volume of the solid object are additive. The volume of the solid object is (see **Figure-23**):

$$V_{solid} = \Delta V = V_2 - V_1$$

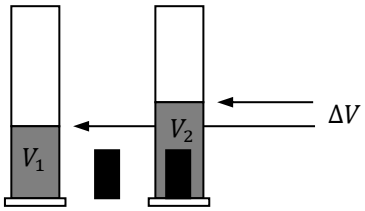


Figure-23: Volume determination by a graduated cylinder.

Measurements can sometimes be difficult to determine. The following are some important techniques. While measuring level of the liquids that have a curve at the surface, volume should be measured from the bottom of the meniscus. The meniscus is the curve formed at the surface of a liquid due to attraction of the liquid for the sides of the container (adhesion).

Measuring from the bottom in **Figure-(24)**, you should get $54 \pm 1\,ml$.

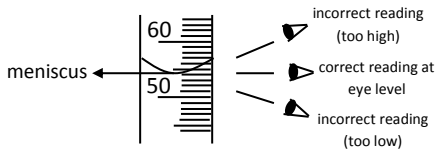


Figure-24: Reading a meniscus. Read the bottom of the minuscus while holding at eye level.

In our case, the size of the graduated cylinder is maximum 250 ml. The scale of the cylinder varies between 50 and 250 ml with an increment of 10 ml. If the volume of a solid is under 50 ml, you should start with at least 50 ml as an initial liquid volume.

Note that, fill the graduated cylinder with a liquid volume in which the measured object is totally sinked. Moreover, be careful about the maximum limit of cylinder for the total (liquid+object) volume.

In general case, uncertainty of the graduated cylinder is taken as 1 division on the scale. However, in our graduated cylinder, the smallest scale is 10 ml and it is a very huge value to use as an error of the measurement. Therefore, it is recommended that a half of the division can be used as an uncertainty of this instrument.

9. Experimental Procedure

In the first part of the experiment, we will try to calculate the densities of the objects with different shapes by measuring their volumes and masses. The volume measurements will be carried out by using four different measuring instruments; ruler, caliper, micrometer and graduated cylinder. Two different device; equal arm balance and digital balance, will be also used to determine the mass of these objects. To get a precise result, three measurements will be done to determine the average dimensions (in order to calculate their volume) and massof each object. The number of trials should be determined by the number of group members. Each group member should take one measurement because making another measurement will be under the effect of the previous measurement. Then, the results of the measurements will be compared with each other by considering their uncertainties. Finally, the average of all volume and mass measurements will be taken in each other and these values will be used to find their common densities. In this calculation, a relation between mass and volume is plotted with taking care of their errors. By analyzing this graph, the density of these objects will be calculated.

9.1. Density

Density, d, is an intrinsic property of materials. In fact, it is independent upon the amount of the object; therefore the density of anything remains the same, no matter the shape and size of the sample. The mass of an object may be distributed in a volume of an object uniformly (being homogeneous) or non-uniformly (being inhomogeneous). If the mass is distributed uniformly throughout the object, the density *d* is defined as the total mass m divided by the total volume *V* of the object. In equation form this is

$$d = \frac{m}{V}$$

9.2. Determination of External Dimensions



Figure-25: A set of objects with different shapes.

In this experiment, the dimensions of the objects will be measured. Therefore, we require the information about how to calculate the volume of the objects by using these measurements.

The **volume** of a rectangular solid is expressed with the equation of

$$\mathrm{V} = \mathrm{W} \times \mathrm{L} \times \mathrm{h}$$

$$\mathrm{V}[cm^3] = \mathrm{Area}[cm^2] \times \mathrm{height}[\mathrm{cm}]$$

where W, L and h are width, length and height, respectively.

Note that, the SI unit of kilogram per cubic meter (kg/m^3) and the CGS unit of gram per cubic centimeter (g/cm^3) are used units for density.

However, g/cm^3 is used commonly to define the density of an object. In general, scientists working in laboratories, dealing with small quantities and distances, prefer to measure distance in centimeters and mass in grams. Moreover, since the cubic centimeter can be alternatively converted to a milliliter or a cc, this fact may affect the usage of CGS system to define density.

SI unit system is based on meter-kilogram-second (MKS) unit set; however, their equivalents in CGS system unit are centimeter-gram-second.

In general, there is a common preference on the MKS system over the CGS system. Although some of the CGS units remain in use for a variety of purposes, their SI units equivalents in the MKS system are used as international standards.

For a sphere, the volume is given by

$$\mathrm{V} = \frac{4}{3}\pi \mathrm{r}^3$$

$$\mathrm{V}[cm^3] = \left(\frac{4}{3}\pi\right) \times \text{cubic radius}[cm^3]$$

where ***r*** is the radius of the sphere.

For a cylinder, the volume is given by

$$\mathrm{V} = \pi \mathrm{r}^2 \mathrm{L}$$

$$\mathrm{V}[cm^3] = \text{Area of the base circle}[cm^2] \times \text{length}[\mathrm{cm}]$$

where ***r*** is the cylinder radius, and ***L*** is its length.

A cone has a circular base and a vertex that is not on the base. Cones are similar in some ways to pyramids. They both have just one base and they converge to a point, the vertex. The formula for the volume of a cone is:

$$\mathrm{V} = \frac{1}{3}\pi \mathrm{r}^2 \mathrm{h}$$

$$\mathrm{V}[cm^3] = \left(\frac{1}{3}\right) \times \text{Area of the base circle}[cm^2] \times \text{height}[\mathrm{cm}]$$

where *r* is the radius of the base circle and *h* is height of the cone from base to the vertex.

Measuring the volume of the objects with ruler

There are two different scales on the ruler that is used in the experiment. One of them is that it has 0.5 mm divisions from 0 to 5 cm. The other is that it has 1 mm divisions from 0 to 15 cm.

If the dimension of the object is less than 5 cm, it is recommended to use first scale which is more sensitive than the other, if not use the main scale which has 1 mm divisions. When measuring the dimensions of the object, try to take the most possible readings considering the parallax. Then, record the result of the measurements for each object to the corresponding tables in the laboratory report.



Figure-26: Ruler.

For the circularly shaped objects, first measure the diameter and then convert it to radius.

Measurements in the cubes and rectangular solids are easier than the others since the ruler is holded parallel to the object.

Be careful to take the each measurement from the different regions of the object.

The height of the cylinder is also straightforward but the diameter of the base circles must be measured from its center. For the holes of the cylinder, ruler is not appropriate to measure its diameter and height (depth of the hole in the cylinder).

To take these measurements, you can benefit from using another object which can be easily inserted in the holes. The length of this object in the hole is equal to the height of the hole. For the cone, you can also use an object to determine its height by ruler. This straight object should be placed on the vertex and parallel to the base.

The height of the cone is equal to the distance between the base and the object on the ruler. For the sphere, again be careful while measuring its diameter as in the case of cylinder.

Measuring the volume of the objects with caliper



Figure-27: Caliper.

1. Loosen the locking screw and move the slider to check if the vernier scale works properly.
2. Check whether the surface of the external, internal and movable jaws, and also depth gauge are clean. If not, clean all these components.
3. Clean the measuring points (or surfaces) of the object which will be contact with the jaws during the measurement.
4. Before making any measurements, determine whether or not the vernier caliper reads zero when the jaws are fully closed. If the caliper does not read zero when the jaws are closed, they are said to have a zero error. Therefore, adjust the caliper's jaws until you get a zero reading. If it is not possible, you must add or subtract the offset from your readings.
5. For external measurements, open the jaws enough to place the **caliper** over the object (Figure-27).
6. For internal measurements, close the internal jaws and placed into object to be measured (**Figure-16**).
7. For the ***depth*** measurements, placed one end of the **caliper** against top of the hole (see Figure-17).
8. Slide lightly the movable jaw to make a contact between the surface of the external, internal or depth gauge with the surface of the object.
9. Tighten the locking screw so that the contact position will not be affected during measurement.
10. Read the main and Vernier scales.
11. Record your results to the corresponding tables in the Laboratory Report.

Measuring the volume of the objects with micrometer



Figure-28: Micrometer.

1. Loosen the lock nut and move the spindle to check if the vernier scale works properly.
2. Clean the anvil and spindle surfaces, and also the measuring object.
3. Before making any measurements, determine whether or not the micrometer reads zero when the anvil and spindle in contact. If the micrometer does not read zero when the anvil and spindle are fully closed, they are said to have a zero error.
4. Rotate the thimble enough to place the micrometer over the object.

5. Place the object in the micrometer so that it is in contact with the measuring surface of the object.

6. Rotate the thimble until the spindle is close to the other side of the object.

7. Turn the ratchet stop until you hear 3 clicks from the ratchet stop.

8. Verify that both the anvil and spindle are touching the object evenly.

9. Set the lock nut while the micrometer is still on the object.

10. Read the sleeve and thimble scales.

11. Record your results to the corresponding tables in the Laboratory Report.

Measuring the volume of the objects with graduated cylinder



Figure-29: Graduated cylinder.

1. Clean the graduated cylinder and the measuring object.

2. Fill the graduated cylinder up to a volume where the object can easily sink, and also this volume should be easily read on the scale of this container. Then, measure it as an initial volume.

3. Place the object in the container, taking care that the liquid level remains above that of the solid. Then, measure the final volume of the combined liquid and solid.

4. Record your results to the corresponding tables in the Laboratory Report.

Report each of the measurements as well as the results with their correct number of significant figures with their corresponding errors.

9.3. Determination of Mass

We can determine the mass of the objects with a laboratory balance by balancing the weight of an unknown mass against the weight of a known mass. Although the balance is between two forces (the weight of the masses), the scales can be calibrated in terms of mass assuming that the force per unit mass is the same for both the known and unknown mass. The unknown mass on a tray at one end of the beam is balanced against the sum of all the known masses placed on the other end. In our system, the slides on the beam will not be used in measurements. They will only be used to calibrate beam with zero reading on a scale. In addition, we can use digital balance as an alternative method to determine the mass of the objects. It will give a digital readout in grams.

Measuring the mass of the objects with equal arm balance



Figure-30: Equal arm balance.

1. Zero the balance before measuring the mass of any object.

   1.1. Be sure all riders are back to the **zero** point.

   1.2. Check to see that the pointer swings freely along the scale.

2. Place the object with an unknown mass in one tray and start adding mass (from a set of mass with different weights) to the other tray until the beam is balanced (parallel with the floor). The pointer must be in zero point if the arms are in equilibrium.

This mass set includes;

Table-**3**: Mass set.

Amount (piece)	Weight
1	500 g
1	200 g
2	100 g
1	50 g
1	20 g
2	10 g
1	5 g
2	2 g
2	1 g
1	500 mg
1	200 mg
1	100 mg
1	50 mg
1	20 mg
1	10 mg



Figure-31: A set of mass with different weights.

3. Record your results to the corresponding tables in the Laboratory Report.

Note that the smallest mass in the set is 10mg; therefore, the uncertainty in this instrument is taken as 10 mg.

Measuring the mass of the objects with digital balance

If you have any digital balance, it can be also used as an alternative method to determine the mass of the objects.



Figure-32: Digital balance.

1. Place the digital balance on a hard and smooth surface as to be parallel position with the floor.

2. Open the balance with pushing the ON/OFF button.

3. Select the appropriate unit of the measurement using MODE button. In addition to gram, there are 3 more measuring units; ounces (oz), pennyweight (dwt) and carat (ct).

1 oz = 28.350 g
1 dwt = 1.555 g
1 ct = 0.2 g

4. Zero the balance before measuring the mass of any object with pushing Z/T button.

5. Place the object with an unknown mass on a platform of the balance, and record the value on the screen.

Note that, the maximum limit of the balance used in this experiment is 200 g.

6. Record your results to the corresponding tables in the Laboratory Report.

While taking measurements, wait until the reading on the screen is stable.

Calculating the density of the objects

The material used to form the cube, rectangular solid, and cylinder and cone shaped objects are the same. In other words, their densities are equal to each other. Therefore, in order to determine this density value,

1. After completing the volume and mass measurements, take the average value of volume and mass for each object.

2. Using each pair of mass and volume values, plot a graph representing the relation between volume and mass.

   Note that, each mass and volume result from one of the objects will form a one data pair.

3. Construct this graph with mass plotted on the *y*-axis and volume plotted on the *x*-axis. Label the axes, and write down the corresponding unit on each axis. Plot the data points along with the error bars.

Note that the distribution of the data points suggests a straight line fitting. The equation of this line will have the familiar form $y = mx + b$, *where* ***m*** *represents the slope of the line, and* ***b*** *represents the* ***y****-intercept.*

4. Draw the best and worst straight lines for your data.

5. Find the slopes of the best line; m_b, and of the worst line, m_w. Calculate the error (uncertainty) in the slope $\Delta m = |m_b - m_w|$.

6. From these slopes, find the density of the unknown material as $d \pm \Delta d$.

The sphere shaped object has a different density as it is observed easily. For this object, again take the average of the results for mass and volume measurements. Then, calculate the density of this object by making proportion between these results.

9.4. Properties of Lenses

A lens is a transmissive optical device. It is typically a thin piece of glass or other transparent material suc as plastic that has the ability to redirect light rays to form an image. It is usually made in circular shape and its two surfaces are ground and polished in a specific usage. A simple lens consists of a single piece of material, while a compound lens consists of several simple lenses. In contrast to mirrors that reflects the incident light, and prisms which refract light without focusing, lenses can be used to focus light to form an image.

Lenses are designed to a desired shape so that to get a variety of types of lenses. Lenses differ from one another in terms of their shape and the materials from which they are made. The two most common types of lenses are concave and convex lenses in which incident rays can converge to a point or appear to be diverging from a point, respectively. These types of lenses are illustrated below in the **Figure-(33)**.

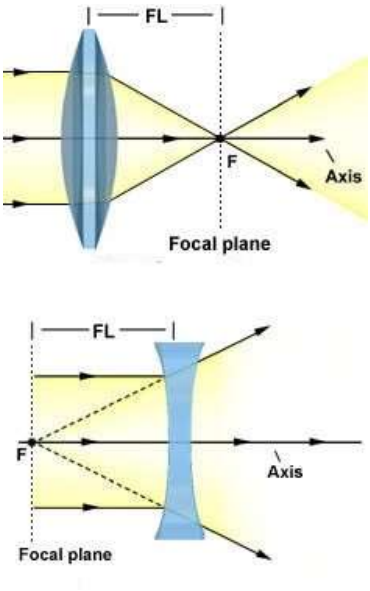


Figure-33: Action of (a) converging lens and (b) diverging lens on incident parallel light rays.

In a simple way, the point at which initially coming light rays meet after passing through a convex lens, or where the the extension of light rays refracted through the concave lens intersect is called a focal point of the lens.

Moreover, the distance between the center of the lens and focal point is defined as focal length. The horizontal line passing throug the center of the lens is called axis of the lens. The point on the surface of lens where the axis meets the lens is known as the vertex. The point in the center of the sphere from which the lens was sliced is known as the center of curvature. The distance from the vertex to the center of curvature is known as the radius of curvature.

Lenses are classified by the curvature of the two optical surfaces.A lens is *bi-convex* (or *double convex*, or just *convex*) if both surfaces are convex. If both surfaces have the same radius of curvature, the lens is *equiconvex*. A lens with two concave surfaces is *biconcave* (or just *concave*). If one of the surfaces is flat, the lens is *plano-convex* or *plano-concave* depending on the curvature of the other surface. A lens with one convex and one concave side is *convex-concave* or *meniscus*. It is this type of lens that is most commonly used in corrective lenses.



Figure-34: Bi-convex lens (a), plano-convex lens (b), concavo-convex lens (c), bi-concave lens (d), plano-concave lens (e) and convexo - concave lens (f).

The defining characteristic of a lens is a quantity called the focal length. Lenses with the same shape and index of refraction will have the same focal length. For different shaped lenses or for a lens in different medium, the focal length should be defined for proper use in optic systems. In other words, to determine whether a lens will behave as a converging or diverging lens based on the curvature of its faces and the relative indices of the lens material, n_1, and the surrounding medium, n_2, the focal length must be known.

For this purpose, the lens-maker's equation gives the reciprocal of the focal length in terms of the physical characteristics of the lens (and the medium in which the lens finds itself);

$$\frac{1}{f} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

where f is the focal length of the lens, n_1 is the index of refraction of the medium surrounding the lens, n_2 is the index of refraction of the material of which the lens is made, R_1 is the radius of curvature of one of the surface of the lens, and R_2 is the radius of curvature of the other surface of the lens. In the application of this equation, the important point is the algebraic sign convention for the R values. Since the convex lens has a *curved out* surfaces, by convention the radius of curvature for a convex surface is positive. On the other hand, with a *curved in* surfaces, in concave lens, by convention, the radius of curvature is negative.

10 Laboratory Report

Name and Surname: _____

Department: _____

Student Number: _____

Date: _____

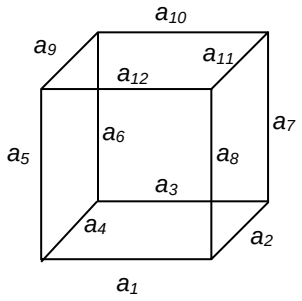
10.1. Measurements of the Object-1.

Data Table-1: Measuring dimensions of the object-1 by ruler (in cm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$a_5 \pm \Delta a_5$	$a_6 \pm \Delta a_6$	$a_7 \pm \Delta a_7$	$a_8 \pm \Delta a_8$	$a_9 \pm \Delta a_9$	$a_{10} \pm \Delta a_{10}$	$a_{11} \pm \Delta a_{11}$	$a_{12} \pm \Delta a_{12}$
1												
2												
3												
Average												

Data Table-2: Measuring dimensions of the object-1 by caliper (in mm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$a_5 \pm \Delta a_5$	$a_6 \pm \Delta a_6$	$a_7 \pm \Delta a_7$	$a_8 \pm \Delta a_8$	$a_9 \pm \Delta a_9$	$a_{10} \pm \Delta a_{10}$	$a_{11} \pm \Delta a_{11}$	$a_{12} \pm \Delta a_{12}$
1												
2												
3												
Average												



From the data in data Table-1, considering the object-1 to be a cube, take the final average of the all average values to determine dimension of "*a"* with its uncertainty. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$ ..... cm

$V \pm \Delta V =$ ..... cm^3

From the data in data Table-2, considering the object-1 to be a cube, take the final average of the all average values to determine dimension of "*a"* with its uncertainty. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$ ..... cm

$V \pm \Delta V =$ ..... cm^3

Data Table-3: Results of the calculation of the volume of the object-1 from dimensional measurements.

	By ruler	By caliper
$V \pm \Delta V$ (cm^3)		

Data Table-4: Measuring the volume of the object-1 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-5: Measuring mass of the object-1.

	By Equal-Arm Balance (in unit of gram)	By digital balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

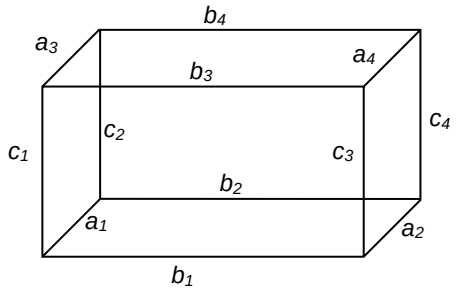
10.2. Measurements of the Object-2.

Data Table-6: Measuring dimensions of the object-2 by ruler (in cm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$b_1 \pm \Delta b_1$	$b_2 \pm \Delta b_2$	$b_3 \pm \Delta b_3$	$b_4 \pm \Delta b_4$	$c_1 \pm \Delta c_1$	$c_2 \pm \Delta c_2$	$c_3 \pm \Delta c_3$	$c_4 \pm \Delta c_4$
1												
2												
3												
Average												

Data Table-7: Measuring dimensions of the object-2 by caliper (in mm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$b_1 \pm \Delta b_1$	$b_2 \pm \Delta b_2$	$b_3 \pm \Delta b_3$	$b_4 \pm \Delta b_4$	$c_1 \pm \Delta c_1$	$c_2 \pm \Delta c_2$	$c_3 \pm \Delta c_3$	$c_4 \pm \Delta c_4$
1												
2												
3												
Average												



From the data in data Table-6, take the final average of the all average values to determine dimension of *a, b* and *c* with their uncertainties. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$..... cm

$b \pm \Delta b =$..... cm

$c \pm \Delta c =$..... cm

$V \pm \Delta V =$..... cm^3

From the data in data Table-7, take the final average of the all average values to determine dimension of *a, b* and *c* with their uncertainties. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$..... cm

$b \pm \Delta b =$..... cm

$c \pm \Delta c =$..... cm

$V \pm \Delta V =$..... cm^3

Data Table-8: Results of the calculation of the volume of the object-2 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V$ (cm^3)		

Data Table-9: Measuring the volume of the object-2 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-10: Measuring mass of the object-2.

	by equal arm balance (in unit of gram)	by digital balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

10.3. Measurements of the Object-3.

Data Table-11: Measuring dimensions of the object-3 by ruler (in cm).

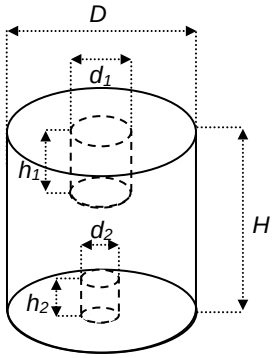
Data Number	$D \pm \Delta D$	$H \pm \Delta H$	$d_1 \pm \Delta d_1$	$h_1 \pm \Delta h_1$	$d_2 \pm \Delta d_2$	$h_2 \pm \Delta h_2$
1						
2						
3						
Average						

From the data in Table-11, determine dimension of *R*, *H*, r_1, h_1, r_2 and h_2 with their uncertainties. Then calculate $V \pm \Delta V$ of the object-3 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3 (volume of the cylinder without the hole)

$V \pm \Delta V =$..... cm^3 (volume of the first hole)

$V \pm \Delta V =$..... cm^3 (volume of the second hole)



$V \pm \Delta V =$..... cm^3 (volume of the object-3)

Data Table-12: Measuring dimensions of the object-3 by caliper (in mm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$	$d_1 \pm \Delta d_1$	$h_1 \pm \Delta h_1$	$d_2 \pm \Delta d_2$	$h_2 \pm \Delta h_2$
1						
2						
3						
Average						

From the data in Table-12, determine dimension of *R, H*, r_1, h_1, r_2 and h_2 with their uncertainties. Then calculate $V \pm \Delta V$ of the object-3 and explicitly show the details of your calculation.

$V \pm \Delta V$ = ... mm^3 = ... cm^3 (volume of the cylinder without the hole)

$V \pm \Delta V$ = ... mm^3 = ... cm^3 (volume of the first hole)

$V \pm \Delta V$ =..... mm^3 = ... cm^3 ((volume of the second hole)

$V \pm \Delta V$ =..... mm^3 = ... cm^3 (volume of the object-3)

Data Table-13: Results of the calculation of the volume of the object-3 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V (cm^3)$		

Data Table-14: Measuring the volume of the object-3 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

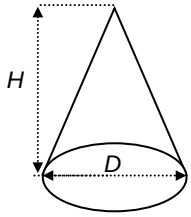
Data Table-15: Measuring mass of the object-3.

	By Equal-Arm Balance (in unit of gram)	By Digital Balance (in unit of gram)
Data Number		
1		
2		
3		
Average		

10.4. Measurements of the Object-4.

Data Table-16: Measuring dimensions of the object-4 by ruler (in cm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$
1		
2		
3		
Average		



From the data in Table-16, determine dimension of R and *H*, with their uncertainties.Then calculate $V \pm \Delta V$ of the object-4 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3

Data Table-17: Measuring dimensions of the object-4 by caliper (in mm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$
1		
2		
3		
Average		

From the data in Table-17, take the final average of the all average values to determine dimension of R and *H* with their uncertainties. Then calculate $V \pm \Delta V$ of the object-4 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3

Data Table-18: Results of the calculation of the volume of the object-4 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V$ (cm^3)		

Data Table-19: Measuring the volume of the object-4 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

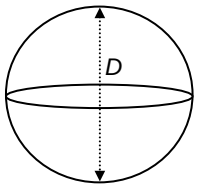
Data Table-20: Measuring mass of the object-4.

	By Equal Arm Balance (in unit of gram)	By Digital Balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

10.5. **Measurements of the Object-5.**

Data Table-21: Measuring dimensions of the object-5 by ruler, caliper and micrometer (in mm).

	by ruler	by caliper	by micrometer
Data Number	$D \pm \Delta D$	$D \pm \Delta D$	$D \pm \Delta D$
1			
2			
3			
Average			



From the data in Table-21, determine dimension of R, with its uncertainty for each measurements. Then calculate $V \pm \Delta V$ of the object-5 and explicitly show the details of your calculations.

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by ruler)

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by caliper)

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by micrometer)

Data Table-22: Results of the calculation of the volume of the object-5 from dimensional measurements.

	by ruler	by caliper	by micrometer
$V \pm \Delta V$ (cm^3)			

Data Table-23: Measuring the volume of the object-5 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-24: Measuring mass of the object-5.

	by Equal Arm Balance (in unit of gram)	by Digital Balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

Determine the **density** of the object-5 from the measured data in **Table-22** and **Table-24** (Show the details of the calculation explicitly).

10.6. Density Calculation

Data Table-25: Summary of the volume measurements for the objects 1-4.

	Volume (cm^3) by Ruler	Volume (cm^3) by Caliper	Volume (cm^3) by Graduated Cylinder	Average value of Volume (cm^3)
Object-1				
Object-2				
Object-3				
Object-4				

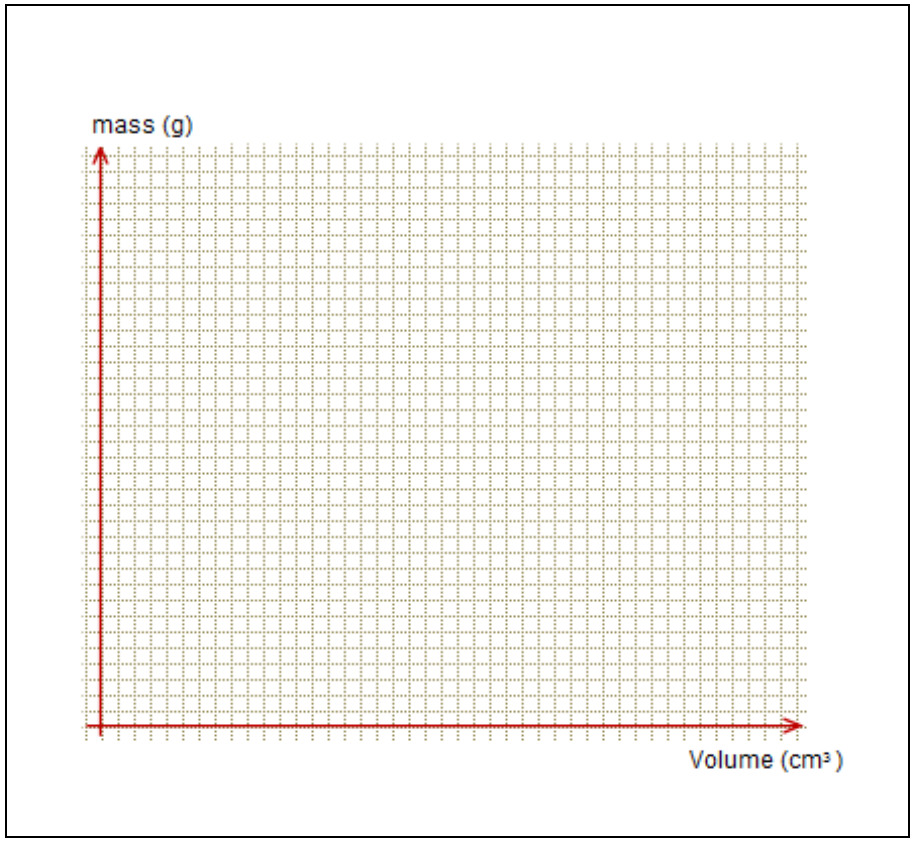
Data Table-26: Summary of the mass measurements for the objects 1-4.

	Mass (g) by Equal Arm Balance	Mass (g) Digital Balance	Average value of Mass (g)
Object-1			
Object-2			
Object-3			
Object-4			

We know that tall these four objects are made from the same material. Then, taking the average volume and mass data set from each independent measurements, plot the graph of the average volume and mass of the objects. For simplicity, put the volume values at the vertical axis, and the mass values at the horizontal axis. Draw the data points along with the corresponding error bars. Noting that the distribution of the data points on the graph paper suggests a straight line fitting, draw the best and worst straight lines. Find the slope $\boldsymbol{m}$ of the best line and the corresponding error $\boldsymbol{\Delta m}$. What are the units of this slope?. What physical quantity does the slope represent?.

$m_b =$ . . . . . .

$m_w =$ . . . . . .



Graph-1: The change in the mass of the objects with respect to the volume.

Report $m \pm \Delta m$ with the correct number of significant figures and units,

$m \pm \Delta m =$ . . . . . .

Phys1103 Experiments

Name:	
Department:	
Student No:	
Date:	

Experiment-2

Straight Line Motion with Constant Velocity

In this section of the experiment, you will study and determine the velocity of an object moving in a straight line with constant velocity. In this type of motion, the moving object travels equal distances in equal time intervals along a straight line. The puck moving freely on the surface of the horizontal (leveled off) air table will be considered as the particle. The puck will move with constant velocity across the air table after it is set in the motion with a small push.

Theory

When a particle moves along a straight line, we can describe its position with respect to an origin (**0**), by means of a coordinate such as x. If there is no net force acting on a moving object, it moves on a straight line with a constant velocity. The displacement of the particle is the change in its position. So, the particle's **average velocity** (v_{av}) during a time interval ($\Delta t=t_2-t_1$) is equal to its displacement ($\Delta x=x_2-x_1$) divided by Δt:

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$ **(Experimental)** (1)

From the Equation-(1), the average velocity is the displacement (Δx) divided by the time interval (Δt) during which the displacement occurs. The direction of the average velocity is always the same as the direction of the displacement. Average velocity does **not** depend on the actual distance that a particle moves (that is, the total distance traveled), but instead **depends on** its original and final position (**displacement**).

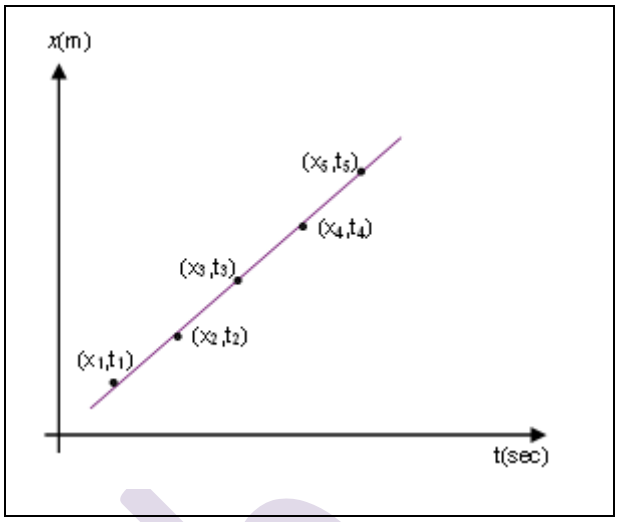


Figure-4: Position as a function of time for an object moving in straight line with constant velocity.

So, the **average velocity** is defined in terms of **displacement**, rather than total distance traveled. The velocity of a particle can also be found from the graph of its **position** as a function of the **time**. On a graph of the **position** (x) versus **time** (**t**), the **average velocity** for a time interval (Δt) is the **slope** of the straight ***line*** connecting the points on the curve that represent the two ends of the time interval. So, in this experiment, if we plot a graph x versus **t** (x-**t** graph), we will have a straight line with a slope and the slope of that line will give the **experimental** value of the average velocity for the motion.

The displacement $x(t)$ of a particle moving in a straight line with constant velocity is given as a function of time in the form:

$$x(t) = x_0 + vt \quad (2)$$

If at **t=0**, the particle was at the **origin**, then $x_0=0$. So, when the object is at the origin with the initial position $x_0=0$, the equation of the motion becomes at any time:

$$x(t) = vt \quad (3)$$

From the above relation, it is obvious that the particle travels equal distance in the equal time intervals along a straight line (see Figure-4).

2.1. Experimental Procedures

Caution!

Do **not** touch the conducting portions, high voltage pins or the hoses of the pucks and the recording or the carbon paper **while spark timer is working**.

1. Construct the experimental set-up as shown in the Figure-(3) without switching on the main power of the **spark-timer**.

   It is noticed that the pucks should be connected to the pump through the hoses. The air hoses should be free and not twisted.

2. Place first the conductive **carbon paper** and then white **recording paper** as the data sheet on the glass plate of the air table.

   2.1. The carbon paper and recording paper should be flat on the air table.
   2.2. Put two pucks on the **recording paper**. **Keep** one of the pucks stationary on a folded piece of data sheet at one corner of the air table. *In this part of the experiment, only **one puck** will be used.*

3. Level the air table **horizontally** by turning on the **compressor** (air supply) and placing one of the pucks at the table's center. Adjust the legs of the air table so that the puck will **not** move.

4. Now, turn on the spark-timer and set the spark timer frequency to:

   - f=20Hz.

5. Test both two switches for the compressor and spark-timer operations. The single puck should move easily, almost without friction when the compressor works only by pressing the **pump** (puck) **foot-switch**. When the **spark-timer foot switch** is pressed, black dots on white paper should be observed (on the side of the recording paper facing the carbon paper).

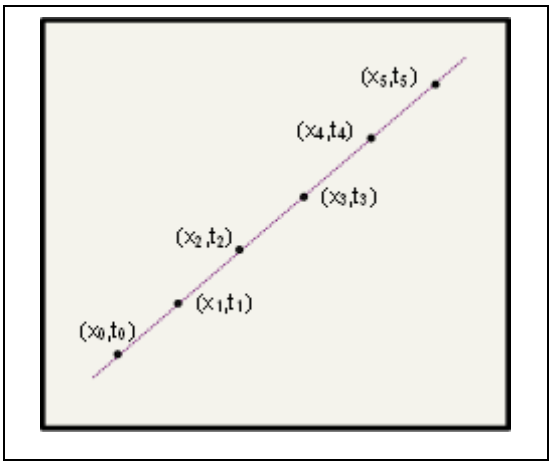


Figure-5: The dots produced by the puck on the data sheet.

6. Then, test the puck motion by activating both the puck **switch** and spark-timer **switch** in the same time and observe the black dots on the recording paper. Make sure that the puck is moving freely on the air table.

7. Activate both compressor (P) and spark timer (S) switches and then, **push** the puck **diagonally** across the surface of the air table.

   7.1. At the moment you **release** the puck, it will move along the whole diagonal distance across the air table in a straight line with **constant velocity**.
   7.2. Then, stop the both switches at the instant the puck reaches its maximum positive displacement, where it will turn back.

8. **Turn off the spark-timer and then remove the puck from the air table for safety.**

9. Remove the white recording paper (data sheet) from the air table.

   9.1. The black dots on the recording paper will look like those given in the Figure-(**5**).
   9.2. The dots made by the spark timer on the recording paper mark the displacement of the puck with equally spaced intervals of **time**.

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10. Number and **encircle** the dots **starting from the first dot** as **0,1,2,….,10**. Take x_0=0 and t_0=0 at the first dot (dot-**0**).

10.1. Using a ruler, **measure** the distances of the **10 dots** starting from **first dot** (**dot-0**).

10.2. The displacement x relative to the initial position (**dot-0**) can be measured directly by using a ruler.

10.3. Find also the **time (t)** corresponding to each dot (that is, consider the **first dot** at time zero and each successive dot as **0.05 seconds** after the previous dot since f=20Hz).

10.4. Record the data values of the position (x) and time (**t**) in the Table-(**1**).

> The time between two dots is 1/20 seconds since the spark timer frequency was set to f=20Hz. So, take the first dot at time zero and each successive dot as "0.05" seconds after the previous dot.

11. Using the data points in Table-(**1**), **plot** the **position** (y-axis) vs **time** (x-axis). The x-**t** graph must show a linear function.

11.1. Label the axes and write down the corresponding **unit** on each axis.

11.2. Draw the best line that fits a linear graph and find the **slope** of the best-line.

11.3. Then, determine the **velocity** (v) of the puck by using the **slope** of the line. The slope of the line is the **average velocity** of the puck.

> The **slope** of the best-fit line on a position-time graph is the average velocity of the object. Average velocity is the rate of change in position for a time interval.

12. Using the data values in Table-(**1**), construct the Table-(**2**).

12.1. Measure the **position** (that is, the **distance** between **2** successive dots) and the **time** corresponding to each **dot interval**.

12.2. Calculate the **average velocity** (v_{av}) from the table for each dot interval.

12.3. Record this on the column marked v_{av} (average speed) in units of **m/sec**.

> Note that the average velocity depends only on the displacement that occurs during the time interval $\Delta t=t_2-t_1$. So, divide each **distance** by the period T=1/f of the spark-timer to obtain the average velocity (v_{av}) of the puck between each pair of dots.

13. **Compare** the **average velocity** for each dot interval with the value which is obtained from the **slope** of graph.

13.1. Fill your data in the Table-(**3**).

13.2. Is the velocity determined by the **slope** consistent with the average velocity calculated for **each dot interval**?. If your comparison shows a percent difference more than 10%, explain at least one of the possible error sources.

Conclusions and Questions

- Discuss the difference in the velocity values calculated for **each dot interval** and the values found from the graph. Is the difference approximately the same?.
- If an object moves at a uniform (that is, constant) velocity during a particular time interval, is its **instantaneous velocity** at any instant the same as its **average velocity**?. Explain your answer.
- Describe how you use a position-time graph to determine average velocity.
- What does a **zero slope** of a position-time graph indicate about the object's average velocity?.
- What is the uncertainty in your measurement of displacements between dots?. Explain briefly the possible sources of the error in the experiment?.

2.2. LABORATORY REPORT

Straight Line Motion with Constant Velocity

Table-1: Data values for the position and time of the motion with **constant velocity**.

Dot Number	Position $x \pm \Delta x(m)$	Time $t(\text{sec.})$	By Slope $v(m/s)$
0	0	0	.....
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Table-2: Experimental data values for the average velocity of the motion with constant velocity.

Interval Number (n)	$x_n(m)$	$x_{n+1}(m)$	$x_{n+1}-x_n(m)$	$t_n(\text{sec.})$	$t_{n+1}(s)$	$t_{n+1}-t_n(s)$	$v_{av}(m/s)$
0-1	**0**	**0**	**0**	**0**	**0**		
1-2							
2-3							
3-4							
4-5							
5-6							
6-7							
7-8							
8-9							
9-10							

(Note that x_n is the position of the nth data point corresponding to the related dot).



Graph-1: The **position** as a function of **time** for the puck moving in a straight line with **constant velocity**.

Table-3: **Comparison** of the velocity found by the slope with the **average velocity** of the each time interval.

Interval Number (n)	$v_{av}(m/s)$	**By Slope** $v(m/s)$	*Difference*, $\lvert\Delta v(m/s)\rvert$	*Difference* (±%)
0-1	. . . . .	. . . . .	. . . . .	. . . . .
1-2	. . . . .		. . . . .	. . . . .
2-3	. . . . .		. . . . .	. . . . .
3-4	. . . . .		. . . . .	. . . . .
4-5	. . . . .		. . . . .	. . . . .
5-6	. . . . .		. . . . .	. . . . .
6-7	. . . . .		. . . . .	. . . . .
7-8	. . . . .		. . . . .	. . . . .
8-9	. . . . .		. . . . .	. . . . .
9-10	. . . . .		. . . . .	. . . . .

Experiment-4 Straight Line Motion with a Constant Acceleration

In this part of the experiment, you will examine straight-line motion of an object (puck) with constant acceleration on an inclined frictionless air table. By plotting the experimental data, you will find the acceleration of the puck sliding down on an inclined air table.

Theory

When a particle slides straight down a frictionless inclined plane, its acceleration is constant, and it will move in a straight line down the plane. The magnitude of the acceleration depends on the angle at which the plane is inclined. If the inclination angle is 90^0, the object will slide down with an acceleration which is equal to the Earth's **gravitational acceleration** g with the magnitude of 9.8 m/s^2.

If you put the puck at the top of the inclined air table and let it slide down the plane, it will move downwards on a straight line but with increasing velocity. The rate of change of the velocity is the acceleration of the puck.

If at time t_1 the puck is at the point x_1 with a velocity of v_1, then at a later time t_2, it will be at a point of x_2 with a velocity v_2. For the motion, the average acceleration of the puck in this time interval $\Delta t = t_2 - t_1$ is defined as:

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \qquad (4)$$

*To investigate the one-dimensional motion of a puck on an air table, we choose a point (dot) near the **beginning** of the motion as the origin of the coordinate system. We determine the location of the puck moving in one dimension by the position x of sparks on the recording paper at **a constant time interval** between the marks on the data sheet.*

Just as **velocity** describes the rate of change of position with time, **acceleration** describes the rate of change of velocity with time. This means that the puck's velocity increases as it goes down the incline.

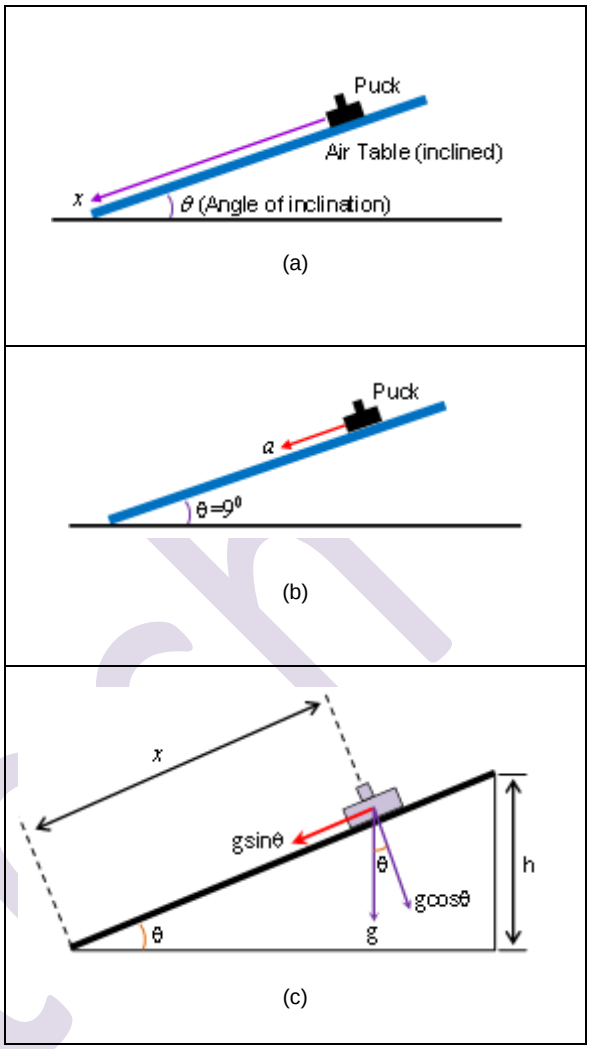


Figure-6: The set-up for a puck moving down an inclined air table (**a**), the straight down motion on an inclined air table (**b**) and components of **g**, the acceleration due to gravity **(c)**. Since the table is inclined at an angle **θ**, the puck's acceleration along the incline should be ***a*=gsinθ**.

If we consider the instantaneous acceleration (simply the acceleration) of the motion in the x-direction, it would be:

$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \qquad (5)$$

In words, the acceleration of an object at any instant is the rate at which its velocity is changing at that instant.

In this experiment, we will observe the motion of a puck moving in a straight line with a velocity changing uniformly. The air table is inclined at an angle of θ with the horizontal plane as shown in the **Figure-(6)**.

The instantaneous acceleration in a straight line motion equals the instantaneous rate of **change of velocity** with time. Since velocity is a vector, acceleration is a vector too. But for one-dimensional motion, we need only use a plus or minus sign to indicate acceleration **direction** relative to a chosen coordinate axis. For the inclined air table, the positive x-axis is taken to be in the **direction** of the puck's motion.

Now, we examine the situation when the acceleration is constant and the motion is in a straight line. In this case, the instantaneous and average accelerations **are equal**. Suppose that at an initial time t_1=0, the puck is at the position of x_0 and has a velocity of v_1=v_0. At a later time t_2=t, it is in position x and has a velocity of v_2=v. If the acceleration of the puck is constant, the **average** acceleration and the **instantaneous** acceleration are **equal** to each other and therefore we find:

$$a = \frac{v - v_0}{t - 0} \qquad (6)$$

Then, the **velocity** of the puck will be:

$$v = v_0 + at \qquad (7)$$

Here, v_0 is the velocity at time t=0 and v is the velocity at any later time t.

For the interval from **t=0** to the later time **t**, the equitation for an object's **motion** with constant acceleration in one dimension can be written as:

$$x = x_0 + v_0 t + \frac{1}{2} a t^2 \qquad (8)$$

where:

x_0 : The displacement at time **t=0** (initial position),

v_0 : The velocity at time **t=0** (initial speed) and,

a : The object's acceleration.

In this experiment, we will allow a puck to accelerate down the incline while the sparktimer is in operation, but stop the sparktimer as the puck hits the bottom part of the inclined plane.

Once a spark (dot) has been selected on the recording paper, we choose a point near the beginning of the track as the **origin** and label it as **x=0** corresponding time **t=0**. Since the motion of object (puck) starts from rest (x_0=0, v_0=0) at **t=0**, the object's position at any time t (instantaneous position) will be:

$$x = \frac{1}{2} a t^2 \qquad \textbf{(Experimental)} \qquad (9)$$

Note that the *positive* x-axis is taken to be in the direction of the puck's motion. Using the displacement (x) from the *origin* to each data point, for each dot; the coordinate x should be measured from the origin to that point.

A graph of Equation-(9), that is, a x-t graph for motion with constant acceleration, is always a parabola that passes through the origin in the x-y plane. However, If a graph of x-t^2 is plotted, we find a straight line which has a **slope** of:

- "$\frac{1}{2}a$"

and it will pass through the origin.

As seen in the Figure-(6c), the acceleration (a) of an object (puck) moving down an inclined plane is a component of the acceleration due to gravity (g). The component of g pointing parallel to the plane causing the acceleration a is:

- $a = g \sin\theta$

- ✓ **So, if θ is known and acceleration (a) is determined, the acceleration due to gravity (g) can be calculated experimentally.**
- ✓ **If θ and g are assumed known, the expected value for the acceleration (a) can be determined.**

4.1. Experimental Procedures

1. Place a sheet of carbon paper and a sheet of recording paper (data sheet) on the air table with the recording paper on top.

2. First, level off the air table and then tilt it to an inclined position by using the block to give an inclination angle (the angle with the horizontal plane) as:
   - $\theta = 9^0$

3. Keep one of the pucks at rest over the carbon paper at the lower right corner of the air table. In this experiment, only one puck will be used.

4. Turn on the compressor (air supply).

   4.1. Put the puck at the top of the inclined plane of the air table and press the compressor pedal (switch).
   4.2. Check if the puck is falling **freely** down the plane.

5. Now, turn on the spark timer and set the spark timer frequency to:
   - f=20Hz.

> Do **not** touch the conducting portions, high voltage pins or the hoses of the pucks and the recording or the carbon paper **while spark timer is working**.

6. Position a puck at the raised end of the inclined air table, close to the middle.

   6.1. Press both puck and spark timer switches simultaneously, release the puck and **stop** pressing when the puck reaches the bottom part of the inclined plane. **Release the puck with no initial velocity**.
   6.2. **Turn off the spark-timer and then remove the puck from the air table.**
   6.3. Remove the data recording paper and examine the dots produced on it.
   6.4. The type of motion that you have observed is the straight line motion with constant acceleration (***acceleration does not change by the time***).

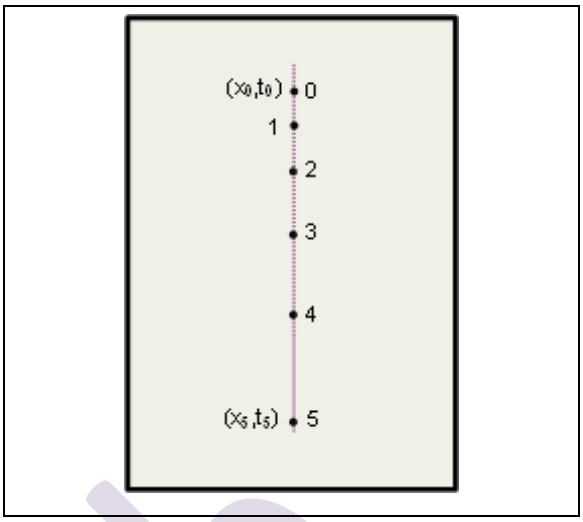


Figure-7: The dots produced by the puck on the data sheet for an inclined air table.

   6.5. Take the trajectory of the puck as the **positive** x-axis (Figure-7).
   6.6. Number and circle the dots (data points**) starting from the first dot** as **0,1,2,…..,10**. Take x**=0** and **t=0** at the first dot as your initial data point.
   6.7. Now, using a ruler, measure the **distance** (**x**) of the remaining dots from the **dot 0**.
   6.8. Also determine and record the **time** (**t**) for each of these dots.
   6.9. Record the data values in the data Table-(4) with respect to your initial point.
   6.10. Calculate and record t^2 values in the Table-(4).

7. **Plot** x versus t^2. Then, draw the best line that fits your data points.

8. Using the slope of this line, determine the **experimental** acceleration (a) of the puck. Show these calculations on your graph paper clearly.

   Note that:
   - $x = \frac{1}{2}at^2$ if $x_0 = 0$.
   - Then, the **slope** of the x versus t^2 graph will be equal to: "$\frac{1}{2}a$".

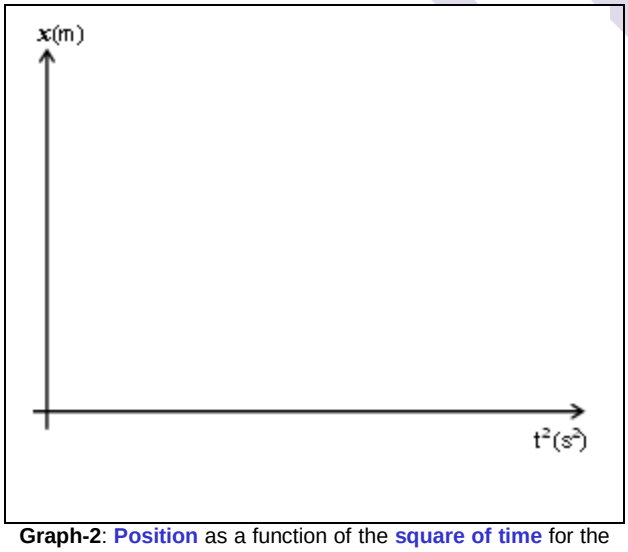
9. Compare your determination of the experimental acceleration (a) with the **expected** value, **gsinθ** where **g**=9.80m/s^2.

4.2. LABORATORY REPORT

Straight Line Motion with a Constant Acceleration

Table-4: Experimental data values for the straight line motion with a **constant acceleration**.

Inclination Angle θ (deg)	Frequency f (Hz)	Dot Number	$x(m)$	$t(\text{sec})$	$t^2(s^2)$	$a(m/s^2)$
			(Measured)			(By Slope)
…..	…..	0	$x_0 = 0$	$t_0 = 0$	0	…..
		1				
		2				
		3				
		4				
		5				
		6				
		7				
		8				
		9				
		10				



Graph-2: **Position** as a function of the **square of time** for the straight line motion with a constant acceleration.

Conclusions and Questions

- Determine the uncertainty in your measurement of displacements between dots on the data sheet?.
- By the slope, what is the **experimental** value of the acceleration (a) for the motion, in units of **m/s^2?**. Is the acceleration of the motion constant?.
- Write the expression for the **expected** acceleration of a particle down a smooth plane, inclined at an angle θ with respect to the horizontal.
- Would choosing a different point as the origin for the analysis of the data **change** the value of the acceleration?. Explain your answer.
- If the acceleration of an object is zero, does it mean that the velocity is zero?. Explain briefly, by giving an example.
- If the angle of inclination θ=20^0 in the **Figure-(6)**, what is the acceleration (a) of the puck (m) if the puck is released from rest at the top of the inclined plane?. Assume **g**=9.8 m/s^2.

Experiment-3 *Projectile Motion*

In this part of the experiment, the puck is projected horizontally with an initial velocity. By the dots produced on the data sheet, we will investigate the motion of the puck along the horizontal and vertical axes independently.

Theory

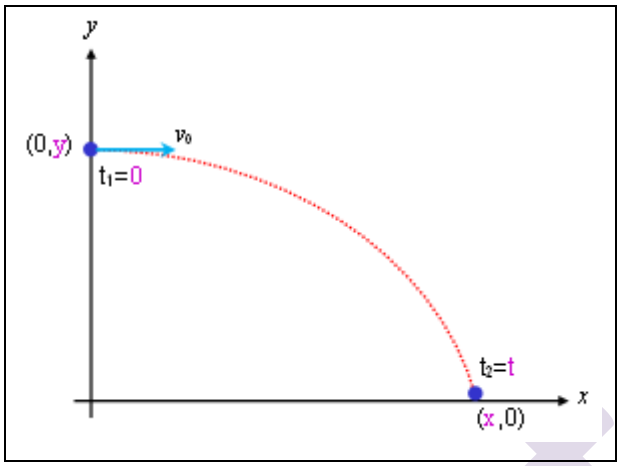


Figure-8: Projectile motion of an object projected horizontally. Trajectory of the object fired horizontally at time $t_1=0$, with the initial speed v_0, from a point at (0; h). The dashed line represents the path of the object.

The other type of the motion in this experiment is the horizontally **projected motion**. Projectile motion is the **two-dimensional** motion of an object under the influence of Earth's gravitational acceleration (***g***). A **projectile** is any object that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration. The **path** followed by a projectile is called its **trajectory**. We consider only its motion *after* it has been projected, and *before* it lands or is caught.

As seen in the **Figure-(8)**, the position of such an object at any given time **t** is given by a set of coordinates x and y which vary with respect to time and represent the horizontal and vertical coordinates, **respectively**.

One of the two components of the **velocity vector** is parallel to the **horizontal** x-axis and the other is parallel to the **vertical**, or y-axis:

$$\vec{v} = v_x\hat{i} + v_y\hat{j} \tag{10}$$

The motion of the object in the horizontal x-direction is a straight line motion with **constant velocity**. In other words, the x-component of the velocity of the object (v_x) will be constant.

However, the acceleration only acts along the vertical direction. The x-component of acceleration is **zero** and y-component is **constant**. This means that only the vertical component of the velocity (v_y) will change with respect to time and the horizontal component of the velocity will be **constant**. Therefore, we can analyze projectile (**two-dimensional**) motion as a combination of horizontal motion with *constant* velocity and the vertical motion with *constant* acceleration.

We can express the vector relationships for the projectile's acceleration by separate equations for the horizontal and vertical components. The components of **the acceleration vector** $\vec{a}$ are:

$$a_x = 0 \tag{11}$$

$$a_y = a \text{ (Constant)} \tag{12}$$

Since there is no acceleration in the horizontal x-direction, the horizontal component of the particle's velocity remains *constant* from its initial value throughout the motion.

In vector form, the acceleration can be expressed as the form:

$$\vec{a} = a_y\hat{j} \tag{13}$$

The position of the object at a given time is given by:

$$\vec{r} = x\hat{i} + y\hat{j} \tag{14}$$

For the two-dimensional (x, y) motion, we can separate acceleration, displacement and velocity in both x and y coordinate directions by the general equations:

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad (15)$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \quad (16)$$

We can write the kinematic equations for constant acceleration (a) in two-dimensional motion as:

x-Component (Horizontal)	y-Component (Horizontal)
$v_x = v_{0x} + a_x t$	$v_y = v_{0y} + a_y t$
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$

Here,

v_x : Horizontal component of velocity,

v_{0x} : **Initial** velocity in the horizontal (x)-direction,

a_x : **Acceleration** in the horizontal (x)-direction,

t : Time for the motion,

x : Horizontal displacement,

x_0 : **Initial** position in the horizontal (x) direction,

v_y : Vertical component of the velocity,

v_{0y} : **Initial** velocity in the vertical (y) direction,

a_y : **Acceleration** in the vertical (y) direction,

y : Vertical displacement,

y_0 : **Initial** position in the vertical (y) direction,

In order to model two-dimensional projectile motion, a metal puck will be set in motion on an inclined **air table**. When the air compressor is switched on, the air is supplied to down the tube under the puck so that it moves with a given initial velocity on the frictionless plane.

Suppose that at time t=0, the particle (puck) is at the point (x_0, y_0) and at this time its velocity components have the initial v_{0x} and v_{0y}. Since the velocity of **horizontal motion** in the projectile motion is constant, we find:

$$a_x = 0 \quad (17)$$

$$\boxed{v_x = v_{0x}} \quad \textbf{(Experimental)} \quad (18)$$

$$x = x_0 + v_x t \quad (19)$$

We can choose the axes of our coordinate system at the point of release of the puck from the shooter at t=0 and take the **downward** direction to be **positive** such that:

$$x_0 = y_0 = 0 \quad (20)$$

This is the origin in the coordinate system associated with the initial position of the puck at t=0. Now, using this relationship in the Equation-(**19**), we will find the equation of motion along the x-axis as:

$$\boxed{x = v_x t} \quad \textbf{(Experimental)} \quad (21)$$

The horizontal displacement for the projectile motion of a small ball projected horizontally is determined by using Equation-(21). If the **horizontal** displacement (x) of the motion and time of flight (t) are measured, then we can find the initial velocity of the object.

For the motion along the y-axis, the velocity (v_y) and **vertical displacement** (y) at the later time (t) become:

$$v_y = at \quad (22)$$

$$\boxed{y = \frac{1}{2}at^2} \quad \textbf{(Experimental)} \quad (23)$$

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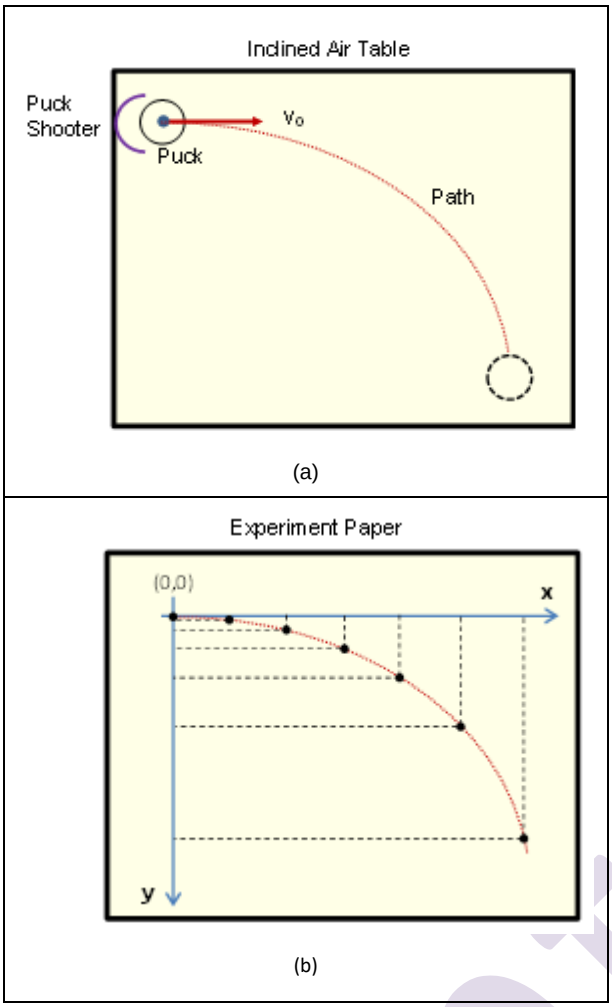


Figure-9: Schematic diagram of the horizontally projected motion of the puck on an inclined air table (a). The data points produced by the puck on the data sheet (b). The ***x*-component** of the velocity of the puck is **constant**. As for motion along the ***y*-axis**, note that the *distance* between the ***y*-projections** of the puck increases with time.

The vertical y-component of the velocity is obtained experimentally and used to determine the acceleration along the y-axis. The projectile motion has a constant horizontal *velocity* and a constant vertical (downward) *acceleration* due to gravity. The vertical distance (y) caused by the change in the velocity is given by Equation-(23).

The dots produced on the data sheet will look like the figure as shown in the Figure-(9). Here, note that the intervals between the dots of the ***x*-projections** in the horizontal direction are equal.

In the projectile motion, the *horizontal* motion and the *vertical* motion are independent of each other, neither motion affects the other. This feature allows you to separate a problem involving two-dimensional motion into two easier one-dimensional problems; one for the horizontal motion (with zero acceleration) and one for the vertical motion (with constant downward acceleration).

We choose our time interval to **begin** (t=0) just as the puck leaves the puck shooter at the position x_0=0, y_0=0. In the horizontal x-direction, the acceleration a_x=0 so the velocity of the puck is constant. In the vertical direction, the distance between the y-projection of the puck increases with time. This situation is identical to the one we have studied earlier, in the case of motion with acceleration (inclined air table). Finally, if you analyze the projectile motion, you can make the following important conclusions:

- The **horizontal** component (***x*-axis**) of a projectile's velocity is constant. (So, the horizontal component of acceleration, in other words, is zero).
- The projectile motion will have a constant **downward** (*y*-axis) **acceleration** due to gravity as seen in the Figure-(**9b**).

3.1. *Experimental Procedures*

In this part of the experiment, you will perform measurements on the motion of a horizontally projected puck on the inclined air table and the motion in two dimensions will be analyzed.

1. First, level off the air table and then bring it into an inclined position with respect to the horizontal by using the block to give an inclination angle:
   - $\theta = 9^0$.

2. Turn on the spark timer and adjust the frequency of the spark timer to:
   - f=20Hz.

3. Keep **one** of the pucks stationary on a folded piece of data sheet paper and carbon paper at the lower corner of the plane.

4. Attach the **shooter** (projectile launcher) to the left side of the inclined plane of the air table about 10cm from the upper edge and **adjust** the shooting angle to 0^0 (zero degrees) **to give horizontal shooting**.

5. First, activate only the compressor switch (P), place the puck into the shooter and make a few test shootings to adjust the best tension of the rubber belt of the shooter to give a convenient trajectory.

6. Now, activate the compressor switch (P) and **place** the puck into the adjusted shooter, then **release** the puck from the shooter **while simultaneously** activating the spark timer (S).

To avoid electrical shock, do not touch the conducting portions of the pucks or the air table while the spark timer is on.

6.1. Stop pressing both pedals when the puck comes to the bottom of the data sheet.

6.2. You will denote these dots of the data points by the **trajectory-B**.

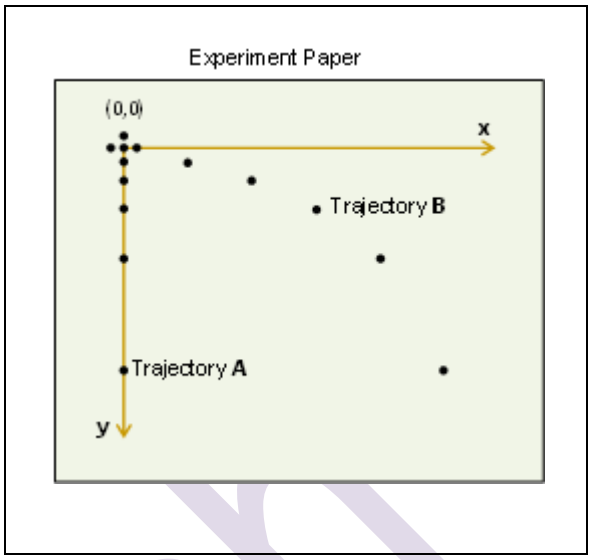


Figure-10: The dots as data points produced by the puck on the data sheet.

7. **On the same data sheet**, **place** the puck opposite to the shooter without tension of shooter (note that the puck must be outside the shooter).

7.1. Then activate both **compressor** and **spark timer** switches in the same time and then **release** the puck from the same initial height, without launching it on the inclined plane.

7.2. This trajectory will be a straight line and the dots will give **trajectory-A**.

8. **Turn off the spark-timer and then remove the pucks from the air table**.

9. Now, **remove the data sheet** to examine the dots of trajectories you have got. You must get the trajectories illustrated in Figure-(10). If the data points are inconvenient to analyze, repeat the experiment and get new data.

10. Select a clear dot on the path at x**=0** and **t=0** as the initial position of the motion.

10.1. Circle and number the data points starting from the first dot as **0, 1, 2, 3, 4, 5,…,10**.

10.2. Consider the downward trajectory as positive y-**axis** and horizontal trajectory as positive x-**axis**.

11. Now, you are ready to analyze projectile motion, horizontally and vertically.

    11.1. First, draw the x and y-axes for **trajectory-B** to determine the **initial position** of the projectile motion.
    11.2. This can be done by first drawing a line parallel to the **trajectory-A** that passes the first dot (**dot 0**) of the **trajectory-B**. This line will give the y**-axis** (take the positive y-direction to be downwards).
    11.3. Then, draw a horizontal line perpendicular to it from **dot 0** to get the x**-axis**.

12. After that, draw from each dot in **trajectory-B** normal to the x and y-axes to get the x and y**-projections** of these dots.

13. Measure the time of flight, t_f (the total time that elapsed during the motion of the puck) and the range x_R of the projectile (the horizontal distance traveled during the motion).

    13.1. Note that the time interval **between two dots** is given by "$1/f$" which is equal to 1/20 seconds.
    13.2. Calculate the projection velocity (*horizontal velocity*) $v_x=v_{0x}$ by using the **time of flight** (t_f) and total **horizontal distance** (x_R) in the Equation-(21).
    13.3. Record your data in the Table-(5).
    13.4. Note that in the horizontal (x) direction of an object projected horizontally, the initial velocity is horizontal and constant (that is, a_x=0).

14. Starting from dot "**0**" of the **trajectory-*B***, measure the distances of the y**-projections** of **10** data points (dots) with respect to the point (origin).

    14.1. Determine also the time (**t**) corresponding to each of these dots.
    14.2. Fill the measurements in the **trajectory-B** columns in the experimental data Table-(**6**).

15. Similarly, by starting from dot "**0**" of the **trajectory-*A***, **measure** the distances (locations) of the first **10** data points (dots) with the corresponding times along the y-axis.

    15.1. For **trajectory-A**, you can calculate the time (**t**) corresponding to each of the dots by using spark-timer frequency.
    15.2. Record your data values in the **trajectory-A** columns of the Table-(6).

10. **Plot** "y" versus "t^2" graphs for both trajectory-A and trajectory-B. Then, draw the best line for the each graph that fits your data points.

    10.1. By using the slope, find the accelerations:
    - a_A and,
    - a_B

    of the **vertical** motions for both trajectory-A and trajectory-B.

    15.3. **Note that** the **slope** of the x versus t^2 graph will be equal to: "$\frac{1}{2}a$".
    15.4. Compare these two accelerations of both trajectories and also compare them with the **acceleration** found in the previous experiment (Experiment-2).
    15.5. Show your calculations with the correct number of significant figures and **units** on your graph paper clearly.

Conclusions and Questions

Path-1 (straight line-trajectory-A):

- Is the **acceleration** constant for the motion along the y-axis?

Path-2 (curved line- trajectory-B):

- Is the **horizontal velocity** of a projected puck constant? Use your graph and data sheet for your explanation.
- If the **velocity** of horizontal motion in the projectile motion is constant, what is the **horizontal acceleration** of a projected puck?.
- Does the **vertical velocity** of the object increase downward in each time interval?.
- Is the **vertical acceleration** of projectile motion constant?.

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3.2. LABORATORY REPORT

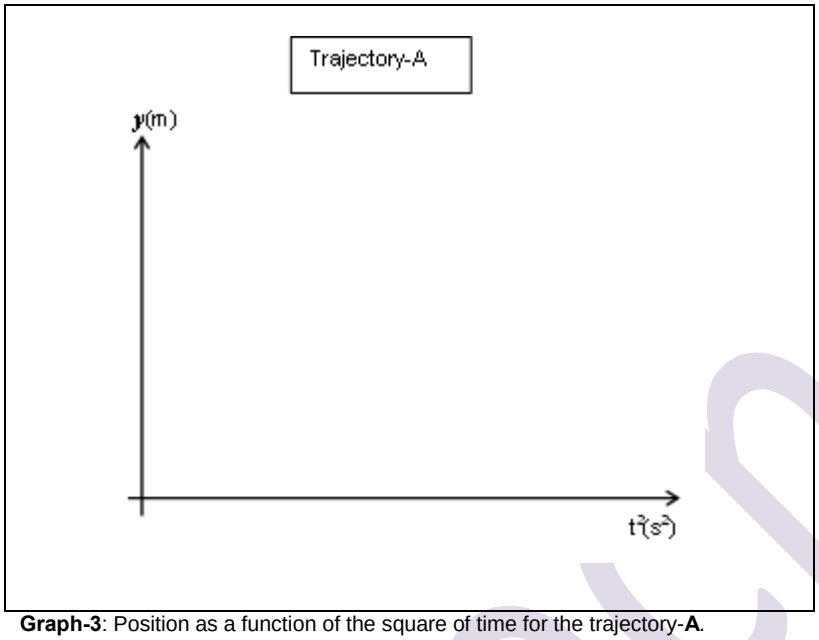
Projectile Motion

Table-5: Horizontal velocity of the **projectile motion**.

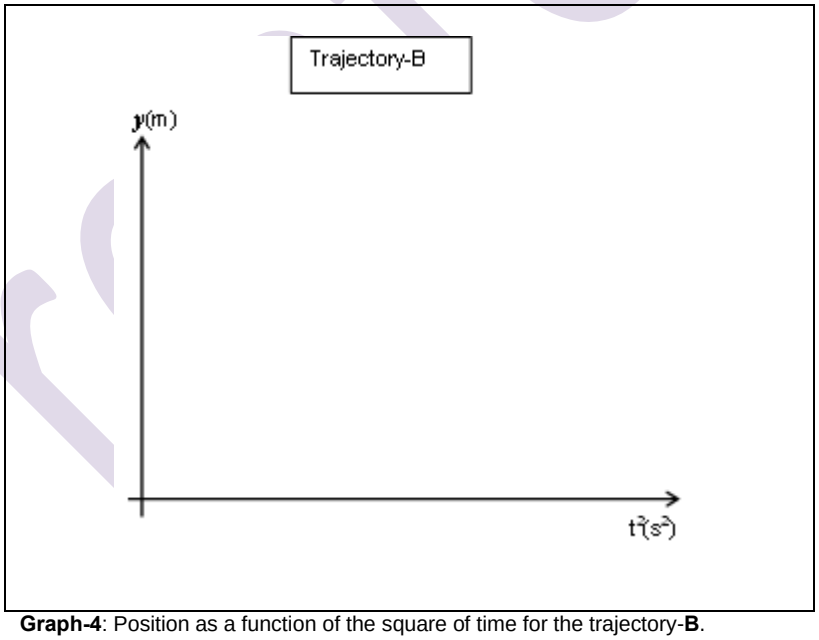
Frequency $f\,(Hz)$	Dot Number	Trajectory-*B* (x -Horizontal Component Motion)				
		$x(m)$	$t\,(\sec)$	$x_R\,(m)$	$t_f\,(\sec)$	$v_x\,(m/s)$
		(Measured)	(Measured)	(Calculated)	(Calculated)	(Calculated)
20	0	0	0	…..	…..	…..
	1					
	2					
	3					
	4					
	5					
	6					
	7					
	8					
	9					
	10					

Table-6: The measurements of accelerations for the trajectory-*A* and trajectory-*B*.

Dot Number	Trajectory-*A*		Trajectory-*B*		Trajectory-*A*	Trajectory-*B*
	(Vertical Motion)		(Vertical Motion)		(By the Slope)	(By the Slope)
	$y\,(m)$	$t\,(\sec)$	$y\,(m)$	$t\,(\sec)$	$a_A\,(m/s^2)$	$a_B\,(m/s^2)$
0	0	0	0	0	…..	…..
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						



Graph-3: Position as a function of the square of time for the trajectory-**A**.



Graph-4: Position as a function of the square of time for the trajectory-**B**.

Experiment-5 Atwood's Machine

In this part, we will use Atwood's machine arranged on an air table to study Newton's second law of motion.

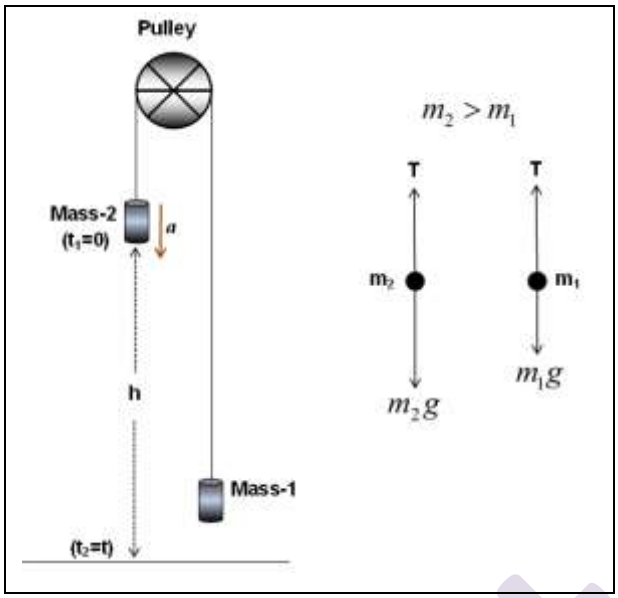


Figure-14: Atwood's machine and free-body diagram of the forces on the each mass.

An Atwood's machine is a physical device used to measure the acceleration of a mass system due to gravity, and to study the dependence of acceleration on the net force as well as on the total mass of a system. In order to move an object that is initially at **rest**, we must apply a force on it. The acceleration of an object depends on the net applied force, and the mass of the object.

The system shown in Figure-(14) is called an Atwood's machine. It consists of two different masses m_1 and m_2 ($m_2 > m_1$) connected at the ends of a string passing over a pulley. The forces acting on each of the masses are also shown in the figure. When the system of two masses is released from the **rest**, the heavier mass m_2 moves downward with **constant** acceleration and the mass m_1 moves upward with the **same acceleration**. Since m_2 accelerates downwards, it experiences a resultant force in this direction such that $m_2 g > T$. As for the mass m_1, we have $m_1 g < T$.

Units in Newton's Second Law

Force	Mass	Acceleration
Newton (N)	Kilogram (kg)	m/s^2

The unit of force is called the **Newton** (**N**). One newton is the amount of net force that gives an acceleration of 1 meter per second squared to a body with a mass of 1 kilogram. So it is related to the units of mass, length, and time (1**N**=1kg.m/s^2).

In the free body diagram of the Atwood's machine; **T** is the tension in the string (cord), m_1 is the **lighter mass**, m_2 is the **heavier mass**, and **g** is the acceleration due to **gravity**. Assuming that there is no friction on the system;

- The net force on m_1 is the difference between the **tension** (T) and $m_1 g$; $T > m_1 g$.
- The net force on the mass m_2 is the difference between the **tension** and $m_2 g$ such that $T < m_2 g$.

The acceleration of an object is directly proportional to the net force acting on it, and is inversely proportional to the object's mass. The direction of the acceleration is in the direction of the net force acting on the object. This is Newton's second law of motion. Newton's second law can be written as the equation:

$$\vec{a} = \frac{\Sigma \vec{F}}{m} \qquad (49)$$

where;

$\vec{a}$:	Acceleration of the object,
m :	Mass of the object,
$\Sigma \vec{F}$:	Net force on the object.

We can apply **Newton's second law** to the bodies of m_1 and m_2 on which the net force is **not** zero. So, the bodies are **not** in equilibrium and hence are accelerating. The net force on the each mass is equal to the mass of the body times its acceleration:

$$\Sigma \vec{F} = m \vec{a} \qquad (50)$$

Newton's second law relates the description of motion (acceleration) to the cause of motion (force). For m_2>m_1, Equation-(50) applied to each mass gives:

$$F_{Net} = T - m_1 g = m_1 a \quad (51)$$

$$F_{Net} = m_2 g - T = m_2 a \quad (52)$$

where;

T :	The tension in the string,
a :	The magnitude of the linear acceleration of **either mass** in the Atwood's machine,
g :	The acceleration due to **gravity**.

The magnitude of the force of the gravity on an object, mg, is called the object's weight.

In an Atwood's machine, the difference in **weight** between two hanging masses determines the **net force** ($\mathbf{F}_{Net}$) acting on the system of both masses. This net force accelerates both of the hanging masses; the **heavier mass** (m_2) is accelerated downward, and the **lighter mass** (m_1) is accelerated upward.

Combining Equations-(51) and (52) leads to:

$$T = m_1 a + m_1 g \quad (53)$$

$$T = m_2 g - m_2 a \quad (54)$$

$$m_1 a + m_1 g = m_2 g - m_2 a \quad (55)$$

$$m_1 a + m_2 a = m_2 g - m_1 g \quad (56)$$

$$\boxed{a = g\left(\frac{m_2 - m_1}{m_1 + m_2}\right)} \quad (57)$$

Equation-(57) gives the theoretical (expected) acceleration for the system of both masses. As seen from the expression, the expected acceleration derived from Newton's second law is equal to "g" times the difference in mass divided by the total mass. So, the net force on the mass system of the Atwood's machine is the difference between the two masses times the acceleration of gravity such that g(m_2-m_1).

If the mass, m_2 is released from rest (v_0=0) at a **vertical** distance h=y above the floor with the **initial** time t_1=0, m_2 strikes the floor at a **final** time t_2=t after its release. The relationship between the distance of the fall (y) and the linear acceleration (a) of the system:

$$y = y_0 + v_0 t + \frac{1}{2} a t^2 \quad (58)$$

Since the initial position of the mass, y_0=0 and the initial velocity, v_0=0, Equation-(58) becomes:

$$\boxed{y = \frac{1}{2} a t^2} \quad \textbf{(Experimental)} \quad (59)$$

The distance (y) through which each mass travels in a time interval (t), starting from rest, is given in terms of the acceleration (a) by the Equation-(59).

Obviously, since the acceleration (a) of the system is **constant** and since both pucks start motion from **rest**, the relation given by the Equation-(59) holds. Solving this expression for the acceleration (a) in terms of the measured quantities of the displacement (y) and the time interval (t) gives:

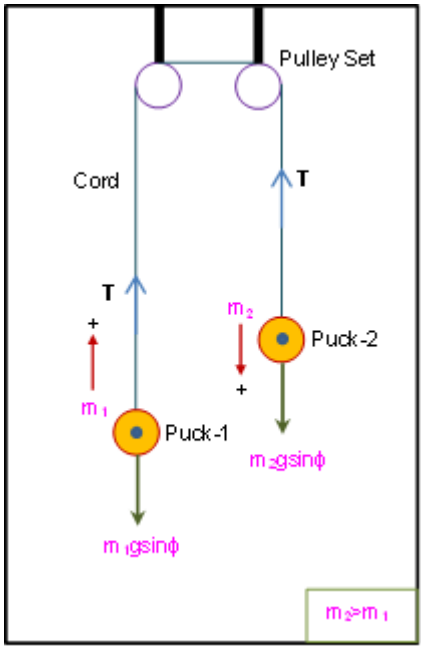
$$\boxed{a = \frac{2y}{t^2}} \quad \textbf{(Experimental)} \quad (60)$$

A graph of **y-t** for the motion with constant acceleration (a) is always a parabola that passes through the origin of that graph. However, if a graph of **y-t^2** is plotted, we will find a straight line which has a **slope** of:

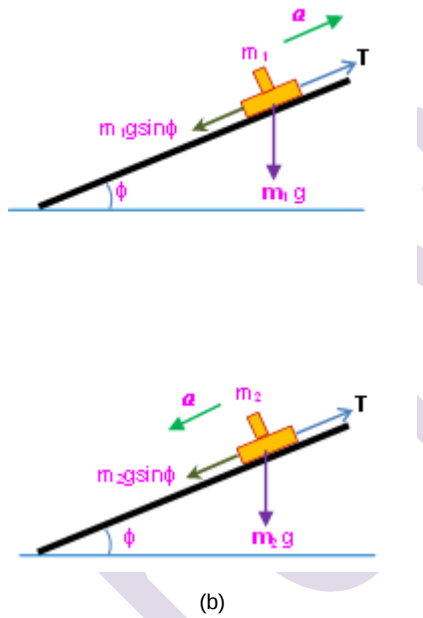
$$\boxed{slope = \frac{1}{2} a} \quad \textbf{(Experimental)} \quad (61)$$

and it will pass through the origin.

Equation-(61) will give the **acceleration** (a) of the Atwood's machine for the applied force of (m_2-m_1)g using a total mass (m_1+ m_2).



(a)



(b)

Figure-15: The experimental set-up of an **Atwood's machine** consisting of two pucks with the masses m_1 and m_2 on an inclined air table (a). The acceleration that is experienced by the both masses is a **constant** with the heavier mass (m_2) falling downwards and the lighter mass (m_1) rising (b).

To construct an Atwood's machine consisting of two pucks connected by a string on an **inclined** air table with **the angle of inclination** of ϕ, we set up the system shown in the Figure-(15a).

The two pucks act as the two masses required for the machine. Putting extra mass on it increases the mass of one of the puck. The greater mass (m_2) is shown in the Figure-(15b). Two forces along the **inclined plane** act on it:

- The tension **T** in the cord pulling upward and,
- The component of its **weight** $m_2 g \sin\phi$.

Since this mass accelerates downward direction, the tension **T** is smaller than $m_2 g \sin\phi$, and therefore the force acting on the mass m_2 is given by:

$$m_2 g \sin\phi - T = m_2 a \qquad (62)$$

Similarly, the net force acting on the mass m_1 is:

$$T - m_1 g \sin\phi = m_1 a \qquad (63)$$

By adding the two equations side by the side and the eliminating, we can find the acceleration:

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g \sin\phi \qquad \textbf{(Expected)} \qquad (64)$$

So, experimental measurement of the distance (***y***) and time (**t**) therefore determines the **acceleration** (***a***), whose value is the result of this experiment. The acceleration (***a***) is related to the acceleration (**g**) due to **gravity** by the Equation-(64).

Note that if the angle of the inclination of the air table is $\phi = 90^0$, then we get:

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g \qquad (65)$$

Solving for the tension in the cord, we can also obtain:

$$T = \frac{2 m_1 m_2}{(m_1 + m_2)} g \sin\phi \qquad \text{(Experimental)} \qquad (66)$$

where, g is the acceleration due to gravity (=9.8m/s^2) and ϕ is the angle that the air table makes with the horizontal (the angle of the inclination).

6.1. *Experimental Procedures*

This part of the experiment is carried out on the inclined air table.

1. First, level the air table off, then put a block under the single leg of the air table by raising its **backside** for an inclined position with the angle:

   - $\phi=20^0$

   You can use an angle finder to measure the inclination angle.

2. Measure the masses m_1 and m_2.

3. Mount the pulley set to the upper side of the inclined air table and place the string (cord) over the both pulleys (see Figure-14a).

   3.1. Arrange the length of the string so that either of the pucks m_1 or m_2 can move at least **20cm** (0,2 meters) on the inclined plane.
   3.2. Remove the air hoses of the pucks and then:
   - Add additional mass to the puck on the right and **record** the total mass of the puck with **extra mass** as m_2.
   - Attach the rings to the two pucks.

   3.3. Connect again the air hoses to the pucks.
   3.4. Then, attach the both pucks to the ends of the **string** passing over the pulleys by using the rings.

   - The distance between the centers of the two pulleys mounted on the upper side of the inclined air table should be **30cm**.
   - The string passing over the pulleys which is attached to the both pucks **should be** parallel to the plane of the air table.

4. Place the puck on the left (with smaller mass, m_1) at the lowest position and the other puck at the highest position.

   Note that the puck, m_2 on the right has extra mass (that is, greater mass).

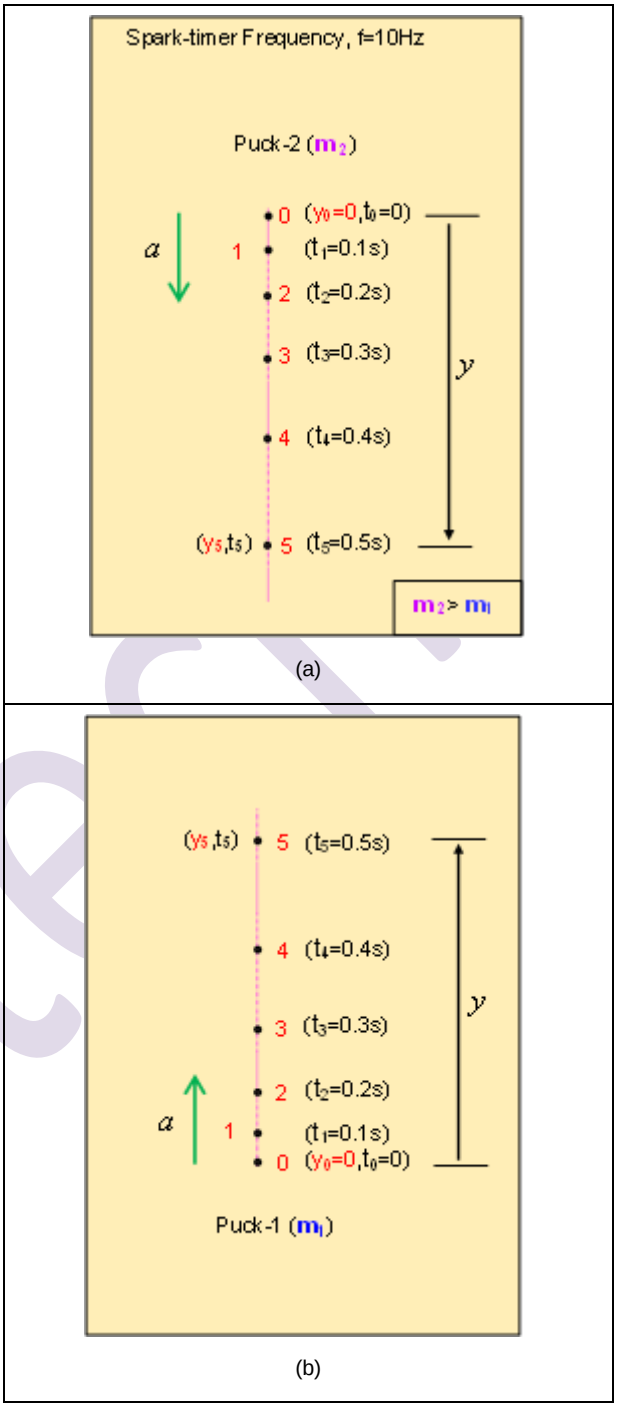


Figure-16: The dots produced by the Puck-**2** (**a**) and Puck-**1** (**b**) on the data sheet in an **Atwood's machine**.

5. Activating **only** the **P**ump, footswitch (P) for the pucks, **release** the system from rest and observe the motion of the two pucks. Repeat this several times to get familiar with the motion.

6. The dots produced on the data sheet will look like those shown in the **Figure-(16)**.

7. Now, set the spark-timer frequency to:

   - f=10Hz.

   which makes a mark *every 1/10 second*. You may, however, change this to 20Hz if your data was not convenient.

8. Repeat now by putting the **two** foot switches on top of each other and depressing them simultaneously to produce the spark record for the both pucks. Observe that the puck on the right goes down while the other goes up the incline.

9. **Turn off the spark-timer and then remove the pucks from the air table.**

10. Remove the data sheet and observe the dots produced on it.

    - ***What kind of a trajectory do the pucks have?.***
    - ***Can you say that both pucks have the same motion?. Explain your answer.***

11. **Starting from the first dot**, number the dots produced by each puck on the data sheet as **0,1,2,3,4, and 5**.

    11.1. The first dot of each trajectory on the data sheet is to be taken as **dot 0**.
    11.2. Use this point (dot **0**) as the reference point for zero position and zero time.
    11.3. Taking the **positive** y-axis to be the direction of motion, **measure** the position and the time of five data points in each trajectory relative to the **dot-0** (0.05 second between successive two dots since f=20Hz).
    11.4. Fill the data values for the position and time in data Table-(8).

Record the position and time measurements of **five** data points of the pocks m_1 and m_2 relative to the **dot 0** in the data Table.

12. Use the data to **plot** the **y** versus t^2 graph for **either** of the masses m_1 or m_2.

The experimental acceleration (a) can be determined by measuring the time (t) required for the mass (m_2) to fall from rest a specific distance (y) as seen in the Figure-16. The acceleration of the each mass is the **same** since they are connected by the string. You should obtain a linear relation between the position (y) and the square of time (t^2) for either of the masses m_1 or m_2. If not, collect the data again with the same settings.

12.1. Now, **draw** the best line passing through the data values on the graph. Be careful with the units of vertical and horizontal axis of the graph.
12.2. Find the **slope** (***s***) of the best-fit line (straight line) and record it in the data Table.
12.3. Find the acceleration (***a***) of the puck from the **slope** of the graph.

13. Calculate experimental value of the **acceleration** (**g**) **due to gravity** (with the Atwood's Machine), in units of **m/s^2**. Compare this with **g**=9.80 m/s^2 to obtain a percentage difference. For this, use:

    - The acceleration (***a***) you obtained,
    - Sinϕ and,
    - The masses m_1 and m_2 in the Equation-(64).

    Show the details of the calculations.

14. Determine the **tension** (**T**) in the cord with the correct unit by using Equation-(66).

15. Find also the **weight** of the mass (m_2). Note that the weight is a force, the pull of gravity acting on the mass of puck (m_2).

16. Show that the **tension** in the cord (string) is smaller than the force (**weight**) along the **inclined plane** since the mass (m_2) accelerates downward.

17. Is your value of the **acceleration** (**g**) consistent with the accepted value of **g**=9.80m/s^2. If your comparison shows a percent difference more than 10%, explain at least one of the possible error sources.

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6.2. LABORATORY REPORT

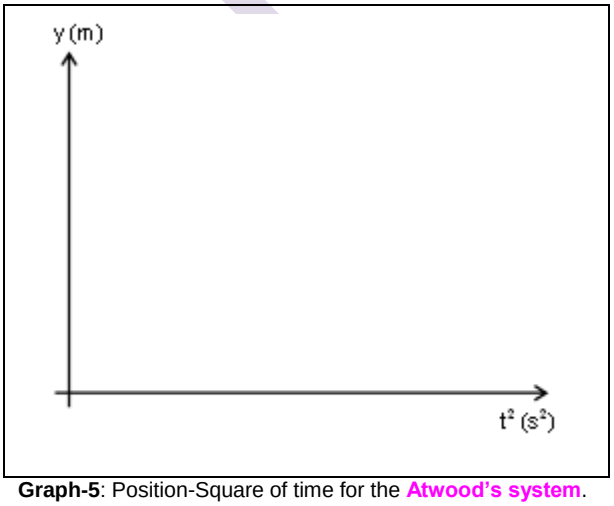
Atwood's Machine

Table-8: Position and time measurements of five data points of the pucks $\mathbf{m_1}$ and $\mathbf{m_2}$ for the **Atwood's system**.

Dot Number	Measured				Experimental
	Position		Time	Square of Time	Slope
	Mass,$\mathbf{m_2}$	Mass,$\mathbf{m_1}$			
	y(m)	y(m)	t(sec)	$t^2(s^2)$	$s(m/s^2)$
0	0	0	0	0	.....
1	.....	.....	.....	.....	
2	.....	.....	.....	.....	
3	.....	.....	.....	.....	
4	.....	.....	.....	.....	
5	.....	.....	.....	.....	

Record the position and time measurements of five data points of the pucks m_1 and m_2 relative to the dot 0.

Angle of inclination, ϕ(deg):	**20**
Spark-timer frequency, f(Hz):	**10**
Puck,m_1(kg):	.....
Puck with extra mass,m_2(kg):	.....
Slope, s(m/s^2):	.....
Acceleration, a(m/s^2):	.....
Acceleration due to gravity, g(m/s^2):	.....
Tension in the cord, T(N):	.....
Weight of the puck with extra mass, W(N):	.....



Graph-5: Position-Square of time for the **Atwood's system**.

Acceleration due to gravity (Experimental)	Acceleration due to gravity (Expected)	Percent Error (Calculated)
g(m/s^2)	g'(m/s^2)	Δg (±%)
.....	.....	.....

Conclusions and Questions

- Describe briefly the basic concept of the Atwood's machine. What is the **net force** which causes an acceleration of the system?.
- How can you determine the **experimental** value of the constant acceleration from the distance and time measurements?. Show your work.
- How can you calculate the **expected** (theoretical) value of the acceleration from the masses and mass differences?.
- What is the relationship between the **acceleration** and the **total mass** when the net force is held constant?.
- If the **net force** due to mass difference remains constant but the **total mass** increases, what happens to the acceleration?.
- How does the acceleration of the **greater mass** (right mass) compare to that of the left mass?.
- Is the acceleration **same** for both masses?. Write a detailed **conclusion** about what you have learned in this experiment.

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Finding static friction coefficient

1. **Purpose**
   - To determine the statical friction coefficients of a mass moving on inclined plane by the effect of gravity.
2. **Theory**

The sum of all the forces acting on a body is called as the net force. If there is a net force on an object, it is accelerated in the direction of this force. In Figure 1, the motion of a mass m on a frictional inclined plane is shown. Due to the gravitational acceleration (W=mg) and the angle of the slope, this force is shown that it has two components. The motion of the mass on frictional inclined plane is due to the x component of this force.

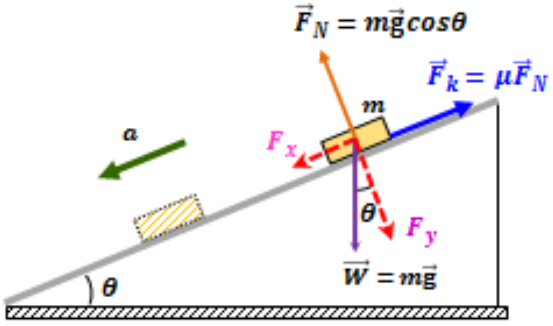


Figure 1. The motion of a mass on inclined plane.

$F_x = mgsin\theta$ - x component

$F_y = mgcos\theta$ - y component

In addition, the y component of the force defines the magnitude of the frictional force owing to the normal force to the mass as formulated below.

$$F_k = \mu F_N$$

The experimental set up is prepared as below.

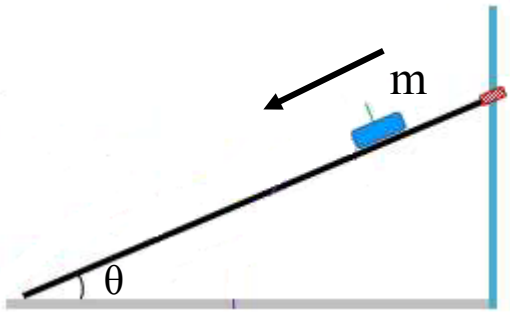


Figure 2. The experimental set-up.

The statical friction coefficient ($\boldsymbol{\mu}$) between the surface of the mass and the surface of the plane will be determined experimentally. To achieve this, the situation of the mass in equilibrium on the plane is examined.

In case of the net force exerted on the mass equals the zero ($\sum \boldsymbol{F_x} = \boldsymbol{0}$), there is a relation between the critical angle ($\boldsymbol{\theta_c}$) where the mass starts to move and statical friction coefficient written below.

$$F_x = F_k$$

$$mgsin\theta = \mu F_N$$

$$mgsin\theta = \mu mgcos\theta$$

$$sin\theta = \mu cos\theta$$

$$\mu = \frac{sin\theta}{cos\theta}$$

$$\boxed{\boldsymbol{\mu = tan\theta}}$$

3. **Experimental Procedure**

- Adjust the board horizontally (θ = 0).
- Place the object at a random position on the board.
- Until the object starts to slide down the board, increase the slope of the inclined plane slowly and record the angle of the board.
- Do this process for three different positions and record the average angle.

$$\boldsymbol{\theta_{avg}} = \frac{\boldsymbol{\theta_1+\theta_2+\theta_3}}{\mathbf{3}}$$

- Calculate the statical friction coefficient according to the average angle.
- Repeat these processes for each surfaces of the object.
- Plot a bar graph of (**μ vs Mass**) and (**μ vs Surface**) for all surfaces.

Table 1. Statical friction coefficients corresponding to the mass and different surfaces.

Surface	**Mass (g)**		**Θ_1 (degree)**	**Θ_2 (degree)**	**Θ_3 (degree)**	**Θ_{avg} (degree)**	**μ**
S_1: Kraft paper	1						
	2						
S_2: Rubber	3						
	4						
S_3: Sand paper	5						
	6						
S_4: Fiberboard	7						
	8						



Figure 3. The bar graph of statical friction coefficients vs masses and surfaces.

Experiment-7 Conservation of Linear Momentum

In this study, we will investigate the conservation of linear momentum of the system of two pucks moving on an air table. The system is isolated from its environment (that is, no external force from an object outside the system causes energy changes in the system). The air table, being horizontal and with the friction almost eliminated, will produce no net external forces on the pucks placed on it. So, we expect the total momentum of the pucks to be conserved.

Theory

The momentum (**p**) of an object is defined as the product of its mass and its velocity:

$$\vec{p} = m\vec{v} \tag{24}$$

The direction of the momentum is the direction of its velocity and the **magnitude** is:

$$p = mv \tag{25}$$

When the net external force on a system of objects is zero, the total momentum remains constant. This is the law of the **conservation of momentum** and the total momentum of an *isolated system* of objects remains constant. The law of conservation of momentum is very useful in dealing with collisions.

If we consider a particle of constant mass (m), we can write Newton's second law for this particle as:

$$\Sigma \vec{F} = m\frac{d\vec{v}}{dt} = \frac{d}{dt}(m\vec{v}) \tag{26}$$

Thus, Newton's second law says that the net force acting on a particle equals to the time rate change of the combination, $m\vec{v}$ (the product of the particle's mass and velocity). This combination is called as the momentum or **linear momentum** of the particle.

If we substitute the definition of momentum into the Equation-(26), we get **Newton's second law** in terms of momentum:

$$\Sigma\vec{F} = \frac{d\vec{p}}{dt} \tag{27}$$

According to Equation-(**27**), the net force (vector sum of all the forces) acting on a particle equals to the time rate change of the particle's momentum. Since momentum is a vector quantity with the same direction as the particle's velocity, we must express the momentum of a particle in terms of its components.

If the particle has the velocity components (v_x, v_y), then its momentum components will be:

$$p_x = mv_x \tag{28}$$

$$p_y = mv_y \tag{29}$$

If there are no external forces (the net external force on a system is zero), the total momentum of the system (the **vector sum** of the momentum of the individual particles that make up the system) is constant or conserved. Each components of the total momentum is separately conserved.

Remember that in **any collision** in which external forces can be neglected, momentum is conserved and the total momentum before equals to the total momentum after. **Only** in **elastic collisions**, the total **kinetic energy** before equals to the total the kinetic energy after. So, in an **elastic collision** between two bodies, the initial and final relative velocities have the same magnitude. The kinetic energy of a particle is a scalar quantity. It depends on only the particle's mass and speed, not its direction of motion.

Collisions in which kinetic energy is not conserved are said to be **inelastic** collisions. The **kinetic energy** that is lost is changed into other forms of energy, often **thermal energy**, so that the total energy is always conserved. Therefore, we do need to remember that even when the kinetic energy is not conserved, the total energy is always conserved.

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For a system with the two pucks, the total momentum (P_t) before the collision will be the same as the total momentum after the collision if the friction can be ignored. By denoting **initial** momentums by the subscript "***i*** " and **final** momentums by the "f ", **the vector equation** for the principle of the conservation of momentum is given by:

$$\vec{P}_{t,i} = \vec{P}_{t,f} \tag{30}$$

$$\vec{P}_{1,i} + \vec{P}_{2,i} = \vec{P}_{1,f} + \vec{P}_{2,f} \tag{31}$$

Let the velocities of the two pucks be denoted as:

- $v_{1,i}$ and $v_{2,i}$ before the collision as the **initial** values and let the velocities after the collision be:
- $v_{1,f}$ and, $v_{2,f}$ as the **final** values.

Then, **the magnitude** of the momentum of the each puck before and after the collision will be:

$$P_{1,i} = mv_{1,i} \tag{32}$$

$$P_{2,i} = mv_{2,i} \tag{33}$$

$$P_{1,f} = mv_{1,f} \tag{34}$$

$$P_{2,f} = mv_{2,f} \tag{35}$$

Now, we can analyze the initial and final velocities of a system with the two particles (puck-A and B). If we define:

$\vec{v}_{1,i}$: The momentum of puck-A, and

$\vec{v}_{2,i}$: The momentum of puck-B just before the collision,

then the total **momentum** of the two pucks before the collisions is the **vector sum**:

$$m_1\vec{v}_{1,i} + m_2\vec{v}_{2,i} \quad \text{(Vector sum)} \tag{36}$$

Immediately after the collision, the pucks have a different velocity and momentum. The total momentum after the collision is the **vector sum**:

$$m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f} \tag{37}$$

The total vector **momentum** of the system of two colliding balls is **conserved** (it stays constant):

Momentum before = Momentum after

$$m_1\vec{v}_{1,i} + m_2\vec{v}_{2,i} = m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f} \tag{38}$$

or,

Momentum before = Momentum after

$$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}'_A + m_B\vec{v}'_B \tag{39}$$

where,

$v_{1,i}$: The velocity of puck-**A** **before** collision (or, v_A),

$v_{2,i}$: The velocity of puck-**B** **before** collision (v_B),

$v_{1,f}$: Final velocity of puck-**A** after collision (v'_A),

$v_{2,f}$: Final velocity of puck-**B** after collision (v'_B).

When the masses of the two pucks are identical ($m=m_1=m_2$), the above relation representing the conservation of the momentum gives the velocity vector relationship:

Momentum before = Momentum after

$$\vec{v}_{1,i} + \vec{v}_{2,i} = \vec{v}_{1,f} + \vec{v}_{2,f} \tag{40}$$

or,

Momentum before = Momentum after

$$\vec{v}_A + \vec{v}_B = \vec{v}'_A + \vec{v}'_B \tag{41}$$

Conservation of the momentum is always valid as long as **no** external forces act. Total momentum is conserved for the two object system in both **elastic** and completely **inelastic collisions**. So, the sum of the initial momentum of the two objects is always equal to the sum of their final momentum.

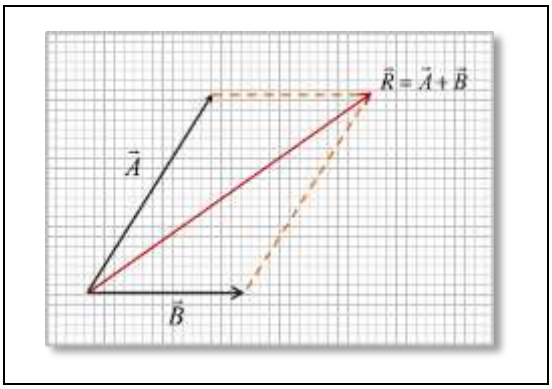


Figure-11: The way to add two vectors by a parallelogram.

When you apply the **conservation of momentum** to a system, remember that momentum is a vector quantity, so you must use **vector addition** to determine the total momentum of the system. To compute the total momentum of the system, we need to add two velocity vectors and to represent the magnitude of the velocity vector. Note that:

- The **magnitude** of a given vector in a scaled vector diagram is depicted by the length of the arrow.
- The arrow is drawn a precise length according to a chosen scale and should have has an obvious arrowhead.

For example, we have **a velocity vector** with a magnitude of 20cm/s. Then, if the scale used for constructing the diagram is chosen to be 1cm=5cm/s, the **vector arrow** of the velocity vector is drawn with a length of $4cm$, such that:

- $4cm\times[(5cm/s)/1cm]=20cm/s$

Therefore, the graphical techniques of vector addition involve drawing accurate scale diagrams to denote individual vectors and their resultants. The parallelogram method is a graphical technique of finding the resultant of two vectors. The way to add two velocity vectors:

- $\vec{A}=\vec{v}_{1,i}$ and,
- $\vec{B}=v_{2,i}$

is to construct a parallelogram as shown geometrically in the Figure-(**11**).

According to the parallelogram law, if two vector quantities are represented by two adjacent sides or a parallelogram, then the diagonal of the parallelogram will be equal to the resultant vector (vector sum), $\vec{R}$ of these two vectors. The resultant is the diagonal of the parallelogram. The magnitude can be determined from the length of its arrow using the scale.

To find the sum of the two vectors geometrically:

1. *Place vectors of:*
   - $\vec{A}$ and $\vec{B}$

   such that their initial points ***coincide****.*
2. *After that, complete a parallelogram that has* $\vec{A}$ *and* $\vec{B}$ *as two sides.*
3. ***The diagonal of the parallelogram*** *with the same initial points as* $\vec{A}$ *and* $\vec{B}$ *is the sum of* $\vec{A}+\vec{B}$.
4. *The* ***vector sum*** $\vec{A}+\vec{B}$ *is called the resultant of vectors* $\vec{A}$ *and* $\vec{B}$.

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In this laboratory, we are also going to investigate the conservation of the kinetic energy of the pucks in the course of the collision.

In an **elastic collision**, the **kinetic energy** of the each colliding object may change, but the total kinetic energy of the system does **not** change. However, in an **inelastic collision** of two objects, the kinetic energy of the two-body system is not conserved. If the objects stick together the collision is a completely inelastic collision and the objects have the same final velocity.

Let's look at an **elastic collision** between two bodies (puck-A and puck-B). To investigate the **conservation of the kinetic energy** of the pucks experimentally, remember the definition of the kinetic energy (K) of an object of mass (m) and linear velocity (v):

$$K = \frac{1}{2}mv^2 \qquad \text{(Kinetic Energy)} \qquad (42)$$

Therefore, the **total kinetic energy** (**K**) of the two-puck system before an **elastic collision** will be:

$$K = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 \qquad \text{Elastic (Before Collision)} \qquad (43)$$

and after the collision, the total kinetic energy:

$$K' = \frac{1}{2}mv_A'^2 + \frac{1}{2}mv_B'^2 \qquad \text{Elastic (After Collision)} \qquad (44)$$

The kinetic energy is conserved in an **elastic collision** such that:

$$\boxed{K = K'} \qquad \text{(Conservation of Kinetic Energy)} \qquad (45)$$

In the completely **inelastic collision**, momentum will obviously be conserved too, since the system is still an isolated one. **But**, the two pucks in the collision will stick together and form a single object with mass **2*m*** that moves with velocity $\vec{v}'$.

The total kinetic energy after the collision will be:

$$K' = \frac{1}{2}(2m)v'^2 = mv'^2 \qquad \text{Inelastic (After Collision)} \qquad (46)$$

In the **elastic collision**, the total kinetic energy should be the same after the collision as it was before the collision. However, it is not conserved in the completely **inelastic collision**. The fractional loss of the kinetic energy is defined as:

$$Fractional\ \ loss = \frac{K - K'}{K} \qquad (47)$$

We can also define the percentage fractional loss in the kinetic energy as:

$$\boxed{\%loss = \frac{K - K'}{K} \times 100} \qquad \text{(Fractional loss)} \qquad (48)$$

We can calculate the percentage fractional loss of the kinetic energy in this experiment by using Equation-(48).

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7.1. Experimental Procedures

This experiment will be carried on a horizontal leveled air table. We will investigate the conservation of momentum for two pucks moving on the frictionless horizontal air table and we will assume there is no net external force on the system. The two pucks will be allowed to collide and then their total momentum before and after the collision will be measured. The details of the experimental procedures are described below.

1. Adjust the legs of the air table for a horizontal leveled air table.

2. Activate only the pump (compressor) switch and push the two pucks diagonally towards each other across the air table.

   2.1. Repeat this procedure several times until you get a satisfactory collision.
   2.2. Don't push the two pucks **too** slowly or too fast, but push them such that they move with a moderate speed.
   2.3. You will find the best speed after several experimental test observations.

3. Now, turn on the spark timer and choose the spark timer frequency as:
   - $f = 20Hz$.

4. Activate the compressor and spark timer switches in the same time, and then push the **two pucks** diagonally to get a collision nearly at the center of the table.

5. Release the two switches when they complete their motion after the collision.

6. **Turn off the spark-timer and then remove the pucks from the air table for safety.**

7. Now, remove the data sheet on the air table to examine the dots produced.

   7.1. The dots of the two pucks should look like those in the Figure-(12a).

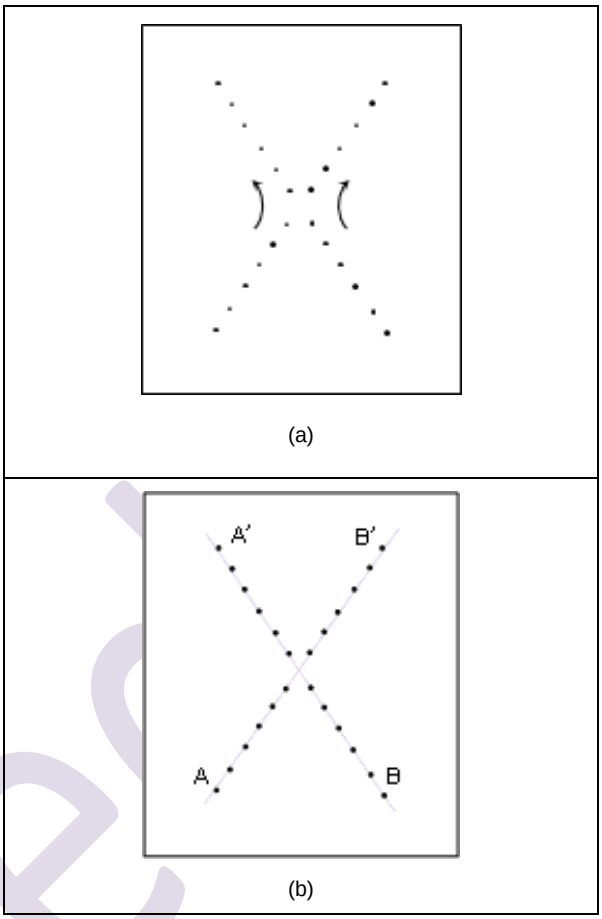


Figure-12: The data points in the elastic collision of the two pucks on a leveled air table (a).The two trajectories of the pucks by **A** and **B** before collision and **A′** and **B′** after the collision (b).

   7.2. Number the dots produced by each puck on the data sheet as **0, 1, 2, 3, 4, 5…10**. You do **not** need to start from the first dot.
   7.3. Label the two trajectories produced by the pucks as **A** and **B** before the collision and then **A′** and **B′** after the collision (Figure-12b).

8. Find the velocity (**v**) of each puck before and after the collision by measuring the length of two or three **intervals** of each trajectory with a ruler and dividing by time.

Since the spark timer frequency is set to f=20Hz, each puck on the table marks 20 dots on the experiment paper in one second, so the time **interval** Δt between two successive dots is given by T=1/20=0.05 second.

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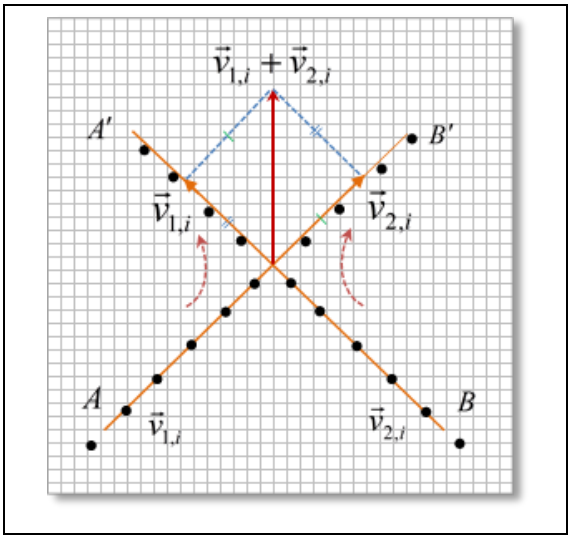


Figure-13: Vector sum given geometrically by constructing a parallelogram.

9. Find the **vector sum** $\vec{v}_{1,i}+\vec{v}_{2,i}$.

9.1. To find vector sum $\vec{v}_{1,i}+\vec{v}_{2,i}$, extend the trajectories **A** and **B** until they intersect on the data sheet as seen in the Figure-(**13**).

9.2. Then, starting from the intersection point, **draw** vectors along the directions of the velocity **vectors** $\vec{v}_{1,i}$ and $\vec{v}_{2,i}$ and with lengths proportional to the **magnitudes** of these velocities $\left|\vec{v}_{1,i}\right|$ and $\left|\vec{v}_{2,i}\right|$.

9.3. You can for instance draw a **1cm** vector to represent a **velocity** of **10cm/s** on your data sheet or on a graph paper having a **scale with 1cm=10cm/s**.

9.4. The **length** of the each vector represents the magnitude and it points in the direction of the velocity.

9.5. As shown in the Figure-(**13**), **add** the two velocity vectors by using parallelogram method **to find** the vector sum (**resultant vector**) of these velocities as:

- $\vec{v}_{1,i}+\vec{v}_{2,i}$ and its magnitude as:
- $\left|\vec{v}_{1,i}+\vec{v}_{2,i}\right|$.

9.6. Record the results in the Table-(7).

10. Repeat the same procedure to find the vector sum of $\vec{v}_{1,f}+\vec{v}_{2,f}$ and the corresponding magnitude.

11. Find the total kinetic energy of the two pucks before and after the collision. Is the kinetic energy conserved?.

7.2. LABORATORY REPORT

Conservation of Linear Momentum

Table-7: Conservation of the linear momentum.

$v_{1,i}\,(m/s)=$	. . . . .
$v_{2,i}\,(m/s)=$	. . . . .
$v_{1,f}\,(m/s)=$	. . . . .
$v_{2,f}\,(m/s)=$	. . . . .
$\left\|\vec{v}_{1,i}+\vec{v}_{2,i}\right\|=$	. . . . .
$\left\|\vec{v}_{1,f}+\vec{v}_{2,f}\right\|=$	. . . . .
$K=$	. . . . .
$K'=$	. . . . .

Conclusions and Questions

- How is the **total momentum** of the system after the collision compared with that before the collision?. Discuss the conservation of the linear momentum by using the data values.
- Write the expression for the total **kinetic energy** of the system both before and after the collision.
- Is the **kinetic energy** of the system conserved in the **elastic collision** of two objects?.
- Is the **kinetic energy** conserved in **completely inelastic collisions**?.
- In the **elastic** or completely **inelastic collisions**, is the **total momentum** of the system always conserved before and after the collisions?.

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Experiment-8 Rotational Motion

To purpose of this experiment is to study the dynamics of a disk that rotates about an axis passing through its center and to calculate the angular acceleration, angular velocity and moment of inertia of that disk. Then we will investigate the conservation of the mechanical energy.

In this experiment we are going to study the rotational motion of a rigid body. A rigid body is an object with a definite shape that doesn't change, so that the particles composing it stay in fixed positions relative to one another. To study the rotational motion of a rigid body, we first need to introduce some new physical quantities and concepts that are essential to the description and understanding of this type of motion.

The first quantity that we are going to introduce is the **moment of inertia** of a rigid body. The moment of inertia (I) of an object is a measure of how much the object *resists* changing its rate of rotation about some axis. We know that in linear motion, if the same force were applied to objects with masses **m** and **M** (**M**>**m**), the one with smaller mass would have greater acceleration. In other words, the force would cause a greater change in the motion of the object with the smaller mass compared to that with the larger mass. So, the mass of an object is actually a measure of the inertia (*resistance*) that an object exhibits against changes in its motion and, the larger this mass is, the larger is the inertia.

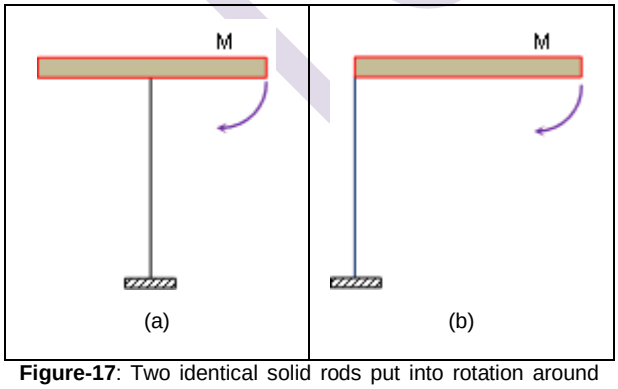


Figure-17: Two identical solid rods put into rotation around two different axes by the same force. It is easier to rotate the rod in (a) than it is in (b).

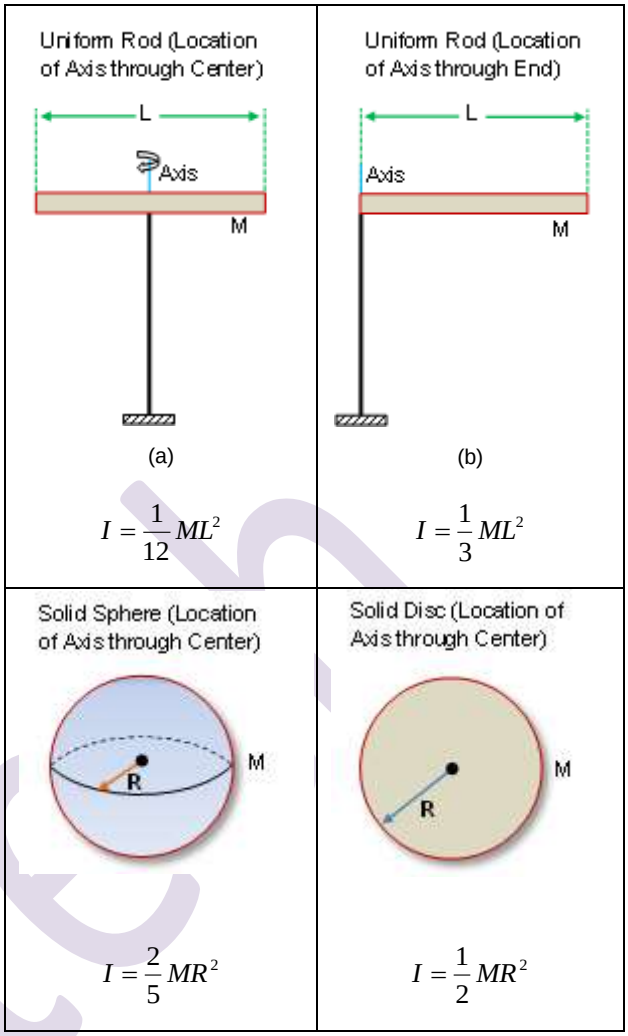


Figure-18: Moment of inertia for various objects of uniform composition.

Consider the situation described in the Figure-(17) where the two uniform rods with the same mass (**M**) are put into two different axes. It is clear that it is harder to rotate the rod in the Figure-(17b). That is, it has a larger moment of inertia compared to the other one, although they both have the same **mass M**. Therefore, the moment of inertia depends on the mass distribution with respect to the axis of rotation in addition to its mass. Figure-(18) gives the moment of inertia for some rigid bodies of uniform composition. Theoretically, for a disk of mass **M** and radius **R**, the moment of inertia through the center of symmetry is:

$$I = \frac{1}{2}MR^2 \qquad (67)$$

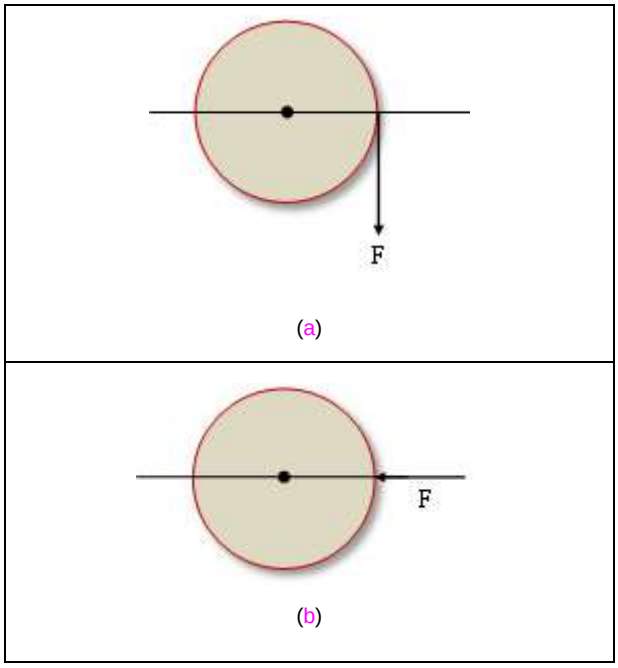


Figure-19: A net force acting on a **disk** that can rotate freely about an axis passing through its center (a). The same force acting as in (b) can not cause rotation.

We know that according to Newton's second law of motion when **a net force** acts on an object that is at rest, it puts it into motion. So, the net force acting tangentially on the disk causes it to rotate about the axis passing through its center. The **net force** (**F**) acting tangentially on the disk in the Figure-(19a) causes it to rotate about the axis passing through its center. However, when the same force acts on the same disk so that its line of action passes through its center as in the Figure-(19b), it can not rotate the disk. We see that a net force is not sufficient to cause a rigid body to rotate. So, we say that a rigid body will experience a change in its rotational motion if a net **torque** of that force about the rotation axis (or point) exists.

The **moment of inertia** of any object about any axis can be determined experimentally, by determining the net **torque** required to give the object an **angular acceleration** (α). The torque (**Γ**) of a force about a point or an axis is defined as:

$$\vec{\Gamma} = \vec{r} \times \vec{F} \qquad (68)$$

Here,

$\vec{r}$: The vector from the axis of rotation to the point where the force is applied,

$\vec{F}$: The applied force.

The torque causes an object to rotate. It depends on the net force and the point at which the force is applied relative to the axis of rotation. By the definition of the vector (cross) product, we have for the magnitude of the **torque**:

$$\Gamma = rF\sin\theta \qquad (69)$$

where θ is the angle between $\vec{r}$ and $\vec{F}$. This expression explains why the net force in the Figure-(19a) causes rotation, while that in the Figure-(19b) does not (that is, the angle θ is zero so Γ=0). Since *torque* is a distance times a force, it is measured in the units of newton-meter (Nm).

Net force acting on an object gives it a net linear acceleration but the torque acting on a rigid body gives it an **angular acceleration** (α) defined by:

$$\vec{\Gamma} = I\vec{\alpha} \qquad (70)$$

where I is the *moment of inertia* of the rigid body about the axis of rotation. The unit of moment of inertia is the kilogram-meter^2 (kgm^2).

As seen from the Equation-(70), to find the moment of inertia (rotational inertia) **experimentally**, a known **torque** (Γ) is applied to the object and the resulting **angular acceleration** (α) of the object is determined.

To describe rotational motion, we make use of ***angular*** quantities, such as angular velocity and angular acceleration. Now, let us define angular velocity (ω) and angular acceleration (α). These angular quantities are defined in analogy to the corresponding quantities in the linear motion and are chosen to describe the rotating object.

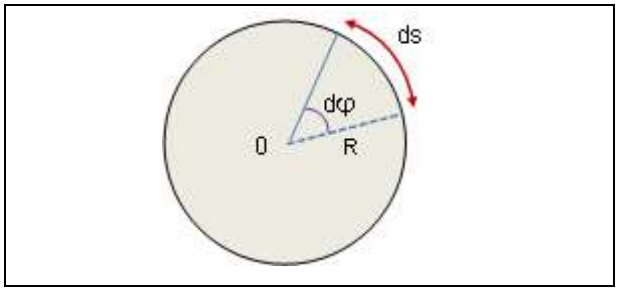


Figure-20: Change in the distance ds and the angle dφ in a time interval dt for a rigid disk rotating about an axis passing through its center.

Consider a rigid disk that is rotating about an axis passing through its center (**Figure-20**). We take the origin of the coordinates to be at the center of the disk. In a time interval **dt**, a point on the rim of the disk will travel a distance ds along the rim of the disk and the radius R will sweep an angle dφ. Each point of a rotating rigid object has, **at any instant**, a linear velocity and a linear acceleration. The magnitude of the **linear velocity** of the point on the rim of the disk will be:

$$v = \frac{ds}{dt} \tag{71}$$

Consequently, the **linear acceleration**, which is the rate of change of the velocity, will be:

$$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2} \tag{72}$$

The **angular velocity** of the rotating disk is defined as the rate of change of the angle φ (measured in radians):

$$\omega = \frac{d\varphi}{dt} \tag{73}$$

Angular velocity has units of radians per seconds (rad/s) and note that all points in a rigid object rotate with the same angular velocity, since every position in the object moves through the same angle in the same time interval.

Angular acceleration is thus:

$$\alpha = \frac{d\omega}{dt} = \frac{d^2 \varphi}{dt^2} \tag{74}$$

Since ω is the same for all points of a rotating object, Equation-(74) tells us that α also will be the same for all points in a rigid body. With "ω" measured in radians per second and "t" in seconds, "α" has units of radians per **second squared** (**rad/s^2**). The length of the arc element *ds* is obviously;

$$ds = Rd\varphi \tag{75}$$

Therefore with ***R*** constant:

$$v = \frac{ds}{dt} = R\frac{d\varphi}{dt} \tag{76}$$

or;

$$\boxed{v = R\omega} \quad \textbf{(Linear Velocity)} \tag{77}$$

Consequently,

$$\boxed{a = R\alpha} \quad \textbf{(Linear Acceleration)} \tag{78}$$

where *R* is a fixed distance from the rotation axis and ω is given in rad/s.

From the above developments, we can relate the linear **quantities** at each point to the angular quantities of the rotating object as:

Linear and Rotational Quantities

Linear Motion	Rotational Motion
v	ω
a	α
m	I
$\vec{F} = m\vec{a}$	$\vec{\Gamma} = I\vec{\alpha}$
$v(t) = v_0 + at$	$\omega(t) = \omega_0 + \alpha t$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

By comparing the expressions for *translational* and *rotational kinetic energies*, we see that angular velocity takes the place of linear velocity and moment of inertia (rotational inertia) takes the place of mass.

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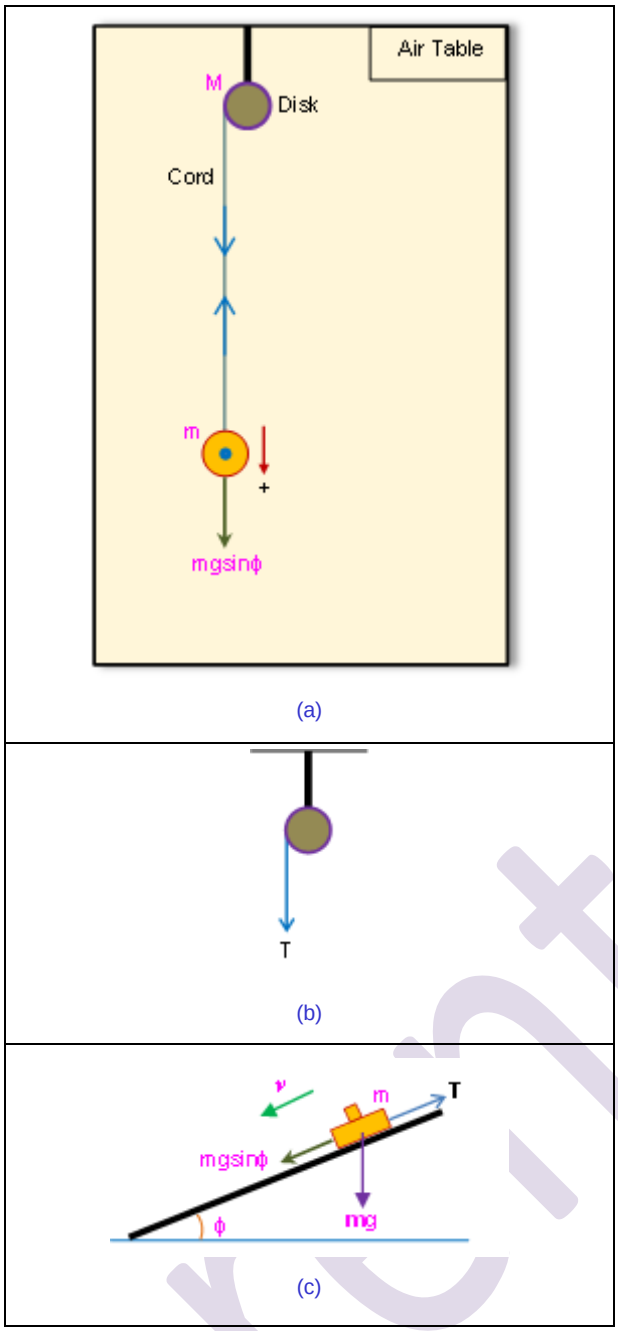


Figure-21: The experimental set-up (**a**). The applied force is the tension (**T**) in the string (**b**). Puck **m** performs linear motion downwards causing the disk **M** to rotate (**c**).

In this experiment, we are going to have the set up shown in the Figure-(21a), above. A solid disk of mass **M** that can rotate freely about an axis passing through its center is mounted to the upper edge of the inclined air table. A cord is wrapped around this disk and the free end of the cord is attached to a puck having the mass (**m**). As the system is released from the rest, the **suspending puck** (**m**) will accelerate down the plane of the inclined air table putting the **disk M** into **rotation**.

The tension **T** in the cord (string) will act as the force whose torque causes the disk to rotate (see Figure-21b). The forces acting on the suspended puck are shown in the Figure-(21c). Applying Newton's second law, we get:

$$mg\sin\phi - T = ma \quad \text{(Experimental)} \quad (79)$$

where ϕ is the angle of inclination of the air table, and a is the linear acceleration of the puck. The angular acceleration (α) of the rotating disk is related to the linear acceleration through:

$$a = R\alpha \quad \text{(Experimental)} \quad (80)$$

where R is the radius of the rotating disk. The torque caused by the tension (T) is given by:

$$\Gamma = RT = I\alpha \quad \text{(Experimental)} \quad (81)$$

where **I** is the **moment of inertia** of the disk. If the system is released from the rest, then at a later time **t**, the linear velocity of the puck will be given by:

$$v = at \quad \text{(Experimental)} \quad (82)$$

Similarly, the angular velocity of the rotating puck will be:

$$\omega = \alpha t \quad \text{(Experimental)} \quad (83)$$

Kinetic energy is the energy associated with the state of motion of an object. The kinetic energy of an object of mass (m) moving with speed (v) is defined as:

$$K = \frac{1}{2}mv^2 \quad \text{(Kinetic Energy)} \quad (84)$$

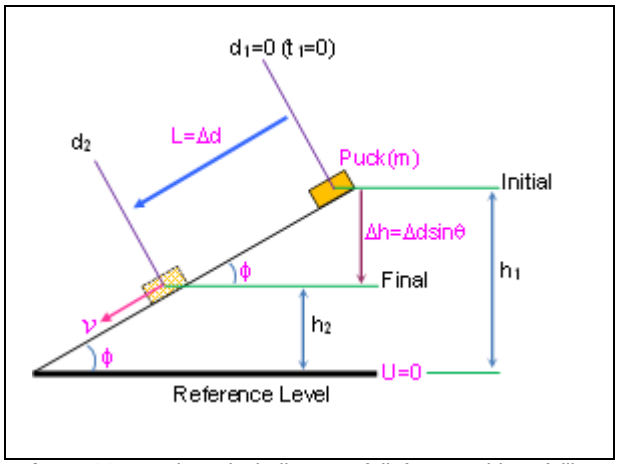


Figure-22: Total vertical distance fall for an object falling down the inclined plane.

The gravitational **potential energy** (**U**) of an object with mass (*m*) at any point of a **vertical height** (*y*) above a chosen reference level can be defined as:

$$U = mgy \qquad \text{(Potential Energy)} \qquad (85)$$

It is important to remember that potential energy is only defined relative to a location (that is, y=0), which can be chosen as the origin of a given coordinate system without affecting the motion of the mass.

As seen in the **Figure-(22)**, if an object of mass (**m**) slides or falls down a total **vertical distance** (**Δ*h***) along an **inclined plane**, its potential energy decreases by:

$$\Delta U = mg(\Delta h) \qquad (86)$$

during its downward motion. The lost potential energy is converted into other types of energy, such as **kinetic energy**. It is noticed that if the object is traveling a **distance** (**L**) along the inclined plane, the total *vertical distance* (**Δ*h***) will be:

$$\Delta h = L\sin\phi \qquad \text{(Vertical Distance)} \qquad (87)$$

where ϕ is the angle of the incline.

In the experiment, the **puck** has kinetic energy due to its translational (**linear**) motion and the **disk** has kinetic energy due to its **rotational** motion. In terms of moment of inertia, the **rotational** kinetic energy of an object rotating about a fixed axis is:

$$K = \frac{1}{2} I\omega^2 \qquad (88)$$

An object rotating about a fixed axis is said to have rotational kinetic energy.

The total kinetic energy of the system **at any instant** of time will thus be:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \qquad \text{(Experimental)} \qquad (89)$$

The initial potential energy is transformed into kinetic energies of the **puck** and **disk**. As the **puck** falls down the inclined plane of the air table, its potential energy changes into **translational** and **rotational** kinetic energies of the **puck** (***m***) and **disk** (**M**), respectively.

Now, we can define a quantity **E**, called the **total mechanical energy** of the system as the sum of the kinetic energy (**K**) plus the potential energy (**U**) of the system at any moment:

$$E = K + U \qquad (90)$$

Letting **U** be the total potential energy of the system, conservation of energy (if the friction neglected) requires that:

$$\Delta K + \Delta U = 0 \qquad (91)$$

$$(K_2 - K_1) + (U_2 - U_1) = 0 \qquad (92)$$

$$\Delta U = -\Delta K \qquad (93)$$

where Δ means the change in energy. If the kinetic energy K increases, then the potential energy U must decrease by an equivalent amount to compensate. Thus the total mechanical energy K+U remains constant when the frictional forces do not act on the objects in the system. This is called the principle of the *conservation of mechanical energy* for conservative forces.

If no friction or non-conservative forces act, we can rewrite **Equation-(92)** as:

$$K_2 + U_2 = K_1 + U_1 \quad (94)$$

So, the quantity **E=K+U** at some time **1** is equal to **K+U** at any time **2**. So, when using the energy approach, we should first define what the initial and final states (the positions and velocities) of the system are. We can use the subscript **1** for initial position and the subscript **2** for the final position.

The **mechanical energy** of a system is the sum of its potential energy and the kinetic energy of the objects within it. Kinetic energy is the energy associated with the state of motion of an object. However, potential energy is only defined relative to a location, which can be chosen as the origin of a given coordinate system without affecting the motion of the mass. When the total mechanical energy of a system is conserved, we can relate the sum of kinetic energy and potential energy at **one instant** to that at **another instant** without considering the intermediate motion.

Due to **conservation of energy**, the **initial** potential energy should be converted completely into kinetic energy, which includes both translational and rotational energies. Therefore, at any instant of time after the system is put into motion, we have:

$$mg(\Delta d)\sin\phi = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \quad \text{(Experimental)} \quad (95)$$

Here, Δd is the change in the distance of the puck travelling down the inclined plane (that is, L=Δd). Before the experiment, you should first determine the system for which you apply energy conservation: the objects and the forces acting.

Equation-(95) is used to verify the conservation of mechanical energy for the system with a puck falling down and a rotating disk.

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8.1. Experimental Procedures

In this part of the experiment, you will use an inclined air table. Therefore, first level off the air table, then bring it into the inclined position by using the block for inclination.

1. First, level off the air table and then tilt it to an inclined position with the angle (ϕ):
   - $\theta = 9^0$.

   by raising its backside and placing the block (the block for inclination) under it.

2. Attach one end of the cord to the rotating disk, **M** on the air table.
   - **2.1.** Attach the other free end of the cord to the suspended puck.
   - **2.2.** As **only** one puck is going to be used in this experiment, keep the other puck still at the lower corner of the air table by folding a piece of paper and putting it under the puck.

3. Adjust the suspended puck so that the rope is stretched, then pressing only the foot switch (P), allow it to fall down the inclined plane.
   - **3.1.** Observe the rotational motion of the disk.
   - **3.2.** Repeat this several times until you get a convenient motion.

4. Now, adjust the suspended puck again as in Step-2, and set the spark-timer frequency to:
   - f=10Hz.
   - **4.1.** Put the foot switches **S** (spark timer switch) and **P** (puck switch) on the top of each other and **press** them simultaneously.
   - **4.2.** Keep the switches depressed until the suspended puck reaches the bottom of the inclined plane.

5. **Turn off the spark-timer and then remove the pucks from the air table.**

6. Remove the data recording paper from the air table and examine the dots produced on it.
   - **6.1.** Is the trajectory of the puck a straight line?.
   - **6.2.** Are the dots evenly (equally) spaced or does the distance between successive dots increase with time?.
   - **6.3.** What kind of trajectory and dot spacing was expected?.
   - **6.4.** Does the data you obtained agree with what you expected?.
   - **6.5.** If you feel that your data is inconvenient, repeat the experiment again.

7. **Starting from the first dot**, number the dots as **0,1,2,3,4, and 5**.
   - **7.1.** Measure the time (**t**) and distance (x) of the each dot from the **dot-0** along with the corresponding errors.
   - **7.2.** Enter your data in the data **Table-(9)**.

8. Using the data in the data Table, **plot** x vs t^2 on a graph.
   - **8.1.** Take the **positive** x-direction in the direction of motion.
   - **8.2.** Now, **draw** the best line passing through the data values on the graph. Be careful with the units of vertical and horizontal axis of the graph.
   - **8.3.** Find the **slope** (s) of the best-fit line (straight line) and from this, determine the **linear acceleration** (a) of the **suspended puck**.
   - **8.4.** Remember that the **slope** of the x versus t^2 graph will be equal to:
     - $"\frac{1}{2}a"$.

   It is noticed that if your graph of **x-t^2** is **not** linear, you should repeat the experiment and again create a new "calculated column" in the data **Table-(9)** that will enable you to obtain a linear relationship.

9. Measure the mass (m) of the suspended puck.

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10. Measure also the **radius** (R) of the **rotating disk** and then using the linear acceleration (a) in the equation:

- $a = R\alpha$

find the **angular acceleration** (α) of the rotating disk.

11. Then, obtain the **linear velocity** (v) of the suspended puck and **angular velocity** (ω) of the rotating puck at the end of the fifth interval (**at t=0.5s**).

12. Now, you will find both the **translational** and **rotational** kinetic energies of the suspending and rotating pucks.

12.1. Using these values of **velocity** (v) and **angular velocity** (ω), find the **total kinetic energy** (K_T) of the system *at this instant* (**at t=0.5s**).

12.2. Then, calculate the potential energy **lost** by the **suspending puck**.

12.3. Verify the **conservation of mechanical** energy for the system. Show your calculations.

12.4. Fill your data in the Table-(10), (11), (12), (13) and (14) with the correct **units**.

13. Using the **mass** (m) of the suspended puck in expression:

- $mg\sin\phi - T = ma$

find the tension (T) in the cord.

13.1. Using the equation:

- $\Gamma = RT$

determine the torque (Γ) of this tension.

13.2. Also find the ***experimental*** value of the moment of inertia (**I**) of the rotating disk by using:

- $\Gamma = I\alpha$

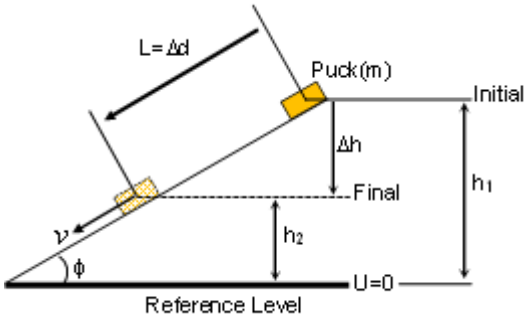
13.3. Now find the **theoretical** value of moment of inertia (**I**) of the disk by:

- $I = \frac{1}{2}MR^2$

13.4. Compare the **experimental** value of the moment of inertia (**I**) with the expected (**theoretical**) value.

13.5. Record you data in the Table-(15), (16) and (17).

14. Consider a puck having mass (**m**) sliding down an inclined plane of the air table as indicated by the diagram below.



Derive the equation for the potential energy **lost** in terms of the inclination angle (ϕ). Suppose the puck is released from rest so that the initial velocity v_0=0 at t=0.

Note that a reference point must be chosen to measure the puck's position vertical to the horizontal plane. For convenience, take h=0 and U=0. The angle (ϕ) of the inclined plane with the horizontal is also shown.

15. Write the expression describing the relationship between the net **torque** (Γ) acting on a rotating disk and its **angular acceleration** (α)?.

16. What is the relationship between the net **torque** acting on the disk and its **radius**?.

17. Write a general expression that represents the relationship between an object's **kinetic energy** due to its translational (linear) motion and its **velocity** (v).

18. Write a statement that gives the total kinetic energy of the mechanical system with a puck due to its **translational motion** and a disk due to its **rotational motion** at any instant of time.

19. Derive the equation describing **conservation of energy** (friction neglected) for an inclined air table system with a puck with linear motion and a disk with rotational motion at any instant of time after the system is put into motion.

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8.2. LABORATORY REPORT

Rotational Motion

Table-9: Position and time measurements of the **suspended puck**.

Spark-Timer Frequency *f* (Hz)	Dot Number	Measured			Experimental	
		Time	Square of Time	Position	Slope	Acceleration
		t(sec)	$t^2(s^2)$	*x*(m)	$s(m/s^2)$	$a(m/s^2)$
10Hz	0	0	0	0	.....	.....
	1					
	2					
	3					
	4					
	5					

Record the time and position measurements of five data points of the suspended puck (m) relative to the dot 0.

Table-10: Translational kinetic energy **gained** by the **suspending puck**.

Suspended Puck		Measured	Calculated	Experimental
Mass	Linear Acceleration $(By\ Slope)$	Time	Linear velocity $(v = at)$	Translational Kinetic Energy $(K_1 = \frac{1}{2}mv^2)$
m(kg)	$a(m/s^2)$	t(s)	v(m/s)	K_1(J)
.....	.....	.....	.....	.....

Translational kinetic energy is determined at the instant of time t=0.5s corresponding to the dot-5 relative to dot-0.

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Table-11: Angular acceleration and velocity of the **rotating disk**.

Rotating Disk		Experimental	Calculated	Experimental
Mass	**Radius**	**Linear Acceleration** $(By\ Slope)$	**Angular Acceleration** $(\alpha = \frac{a}{R})$	**Angular Velocity** $(\omega = \alpha t)$
M(kg)	R(m)	a(m/s^2)	α (rad/s^2)	ω (rad/s)
.....	.....	.....	.....	.....

Table-12: Rotational kinetic energy **gained** by the **rotating disk**.

Rotating Disk		Calculated	Calculated	Experimental
Mass	**Radius**	**Moment of Inertia** $(I = \frac{1}{2}MR^2)$	**Angular Velocity** $(\omega = \alpha t)$	**Rotational Kinetic Energy** $(K_2 = \frac{1}{2}I\omega^2)$
M(kg)	R(m)	I(kgm^2)	ω (rad/s)	K_2(J)
.....	.....	.....	.....	.....

Table-13: Total kinetic energy of the system **gained** by the **suspending puck (m)** and **rotating disk (M)**.

Suspended Puck	Rotating Disk	Experimental
Translational Kinetic Energy $(K_1 = \frac{1}{2}mv^2)$	**Rotational Kinetic Energy** $(K_2 = \frac{1}{2}I\omega^2)$	**Total Kinetic Energy** $(K_T = K_1 + K_2)$
K_1(J)	K_2(J)	K_T(J)
.....	.....	.....

Table-14: Potential energy **lost** by the **suspending puck**.

Puck	Acceleration due to Gravity	Measured	Measured	Experimental
		Change in the Distance	**Angle of the Incline**	**Potential Energy Lost** $(\Delta U = mg\Delta d\sin\phi)$
m(kg)	g(m/s^2)	Δd(m)	ϕ(deg)	ΔU(J)
.....	.....	.....	.....	.....

Table-15: Tension in the cord.

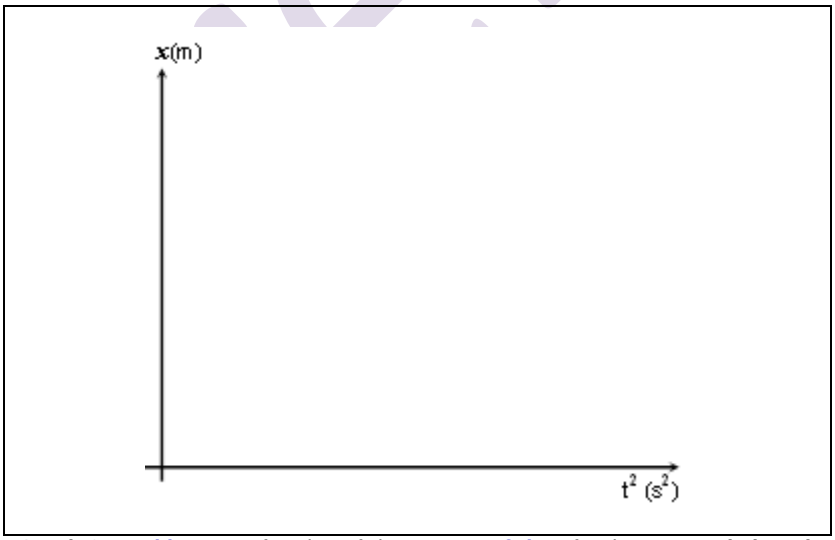
Puck	Acceleration due to Gravity	Measured	Experimental	Experimental
		Angle of the Incline	Acceleration ($By\ Slope$)	Tension ($mg\sin\phi - T = ma$)
m(kg)	g(m/s^2)	ϕ(deg)	a(m/s^2)	T(N)
.....	.....	.....	.....	.....

Table-16: Torque caused by the tension.

Rotating Disk	Experimental	Experimental
Radius	Tension ($mg\sin\phi - T = ma$)	Torque of the Tension ($\Gamma = RT$)
R(m)	T(N)	Γ(Nm)
.....	.....	.....

Table-17: Moment of Inertia of the disk.

Rotating Disk	Calculated	Experimental	Expected	Difference
Torque of the Tension ($\Gamma = RT$)	Angular Acceleration ($\alpha = \frac{a}{R}$)	Moment of Inertia ($\Gamma = I\alpha$)	Moment of Inertia ($I = \frac{1}{2}MR^2$)	Moment of Inertia ($\Delta I = I' - I$)
Γ(Nm)	α (rad/s^2)	I(kgm^2)	I′(kgm^2)	ΔI(kgm^2)
.....	.....	.....	.....	.....



Graph-6: Position as a function of the square of time for the **suspended puck** moving down the plane of the inclined air table with the linear acceleration.